

Financial Statement Analysis

Solution Manual

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Chapter 1 — A Framework for Financial Statement Analysis

Section A — Numerical Exercises

Exercise 1 — Identifying the Analytical Impact of an Accounting Change

a) Depreciation under old policy (10 years): $\text{€}600\text{m} / 10 = \text{€}60$ million per year.

b) Depreciation under new policy (15 years): $\text{€}600\text{m} / 15 = \text{€}40$ million per year.

c) Reduction in depreciation due to policy change: $\text{€}60\text{m} - \text{€}40\text{m} = \text{€}20$ million.

Impact on EBIT: $+\text{€}20$ million (EBIT is overstated by $\text{€}20\text{m}$ relative to old policy).

Impact on Net Income: $+\text{€}20\text{m} \times (1 - 0.30) = +\text{€}14$ million.

d) Restated Income Statement (on like-for-like basis):

Revenue:	€800 million (unchanged)
Operating Costs (excl. D&A):	€550 million (unchanged)
Depreciation (restated at €60m):	€60 million
EBIT (restated):	€190 million
Net Income (restated):	€106 million

Note:

Reported EBIT = $800 - 550 - 40 = \text{€}210$ million.

Restated EBIT = $800 - 550 - 60 = \text{€}190$ million.

Reported Net Income = $\text{€}120$ million.

Restated Net Income = $\text{€}120\text{m} - \text{€}14\text{m} = \text{€}106$ million.

INTERPRETATION: The reported EBIT of $\text{€}210\text{m}$ and net income of $\text{€}120\text{m}$ are inflated by $\text{€}20\text{m}$ and $\text{€}14\text{m}$ respectively due to an accounting policy change that is entirely discretionary. From an equity analyst's perspective, the change has no economic substance — the assets do not become more durable simply because management decides to depreciate them more slowly. The analyst should use the restated figures for valuation purposes and treat the accounting change as a red flag regarding the quality of reported earnings.

Exercise 2 — Assessing the Impact of Consolidation Scope on Reported Performance

a) Organic growth computation:

Reported FY2023 Revenue: $\text{€}1,500$ million

Less: Acquired revenue (Delta H2 2023): $-\text{€}180$ million

Organic FY2023 Revenue: $\text{€}1,320$ million

Adjusted FY2022 Revenue (excl. divested unit): €1,200m – €60m = €1,140 million

Organic Revenue Growth = $(€1,320\text{m} - €1,140\text{m}) / €1,140\text{m} = €180\text{m} / €1,140\text{m} = +15.8\%$

b) Organic growth isolates the performance of the company's existing business — the commercial engine that management controls and that can be expected to continue generating revenue in future periods. Acquired revenues are one-time step-changes that inflate the year-on-year comparison without reflecting an improvement in the underlying business. For valuation purposes, the analyst needs to understand how fast the 'like-for-like' business is growing, because this is what drives the terminal value in a discounted cash flow model.

c) Complexity factor: Group of companies / scope of consolidation. Mergers and acquisitions and divestitures change the composition of the reporting entity, making year-on-year comparisons of headline figures inherently misleading. This is a standard complexity/risk factor that analysts must always screen for before interpreting revenue growth trends.

Exercise 3 — Cash vs. Accrual: Diagnosing the Quality of Reported Earnings

a) OCF / Net Income:

FY2021: $75/80 = 93.8\%$

FY2022: $60/95 = 63.2\%$

FY2023: $30/115 = 26.1\%$

Trend: The ratio is falling sharply. In FY2021, approximately 94 cents of every euro of net income was backed by cash. By FY2023, only 26 cents is. This is a serious warning sign. Earnings are growing on paper, but cash generation is collapsing.

b) Days Sales Outstanding (DSO):

FY2021: $(90/500) \times 365 = 65.7$ days

FY2022: $(130/560) \times 365 = 84.7$ days

FY2023: $(200/640) \times 365 = 114.1$ days

Trend: DSO has increased by nearly 50 days over three years. Customers are taking much longer to pay — or the company is recording revenue before it is genuinely earned/collectible (aggressive revenue recognition).

c) Credit analyst perspective:

RED FLAGS: (i) Earnings growth with collapsing cash conversion is a classic signature of earnings manipulation — revenues may be recognized prematurely or receivables may not be fully collectible. (ii) Rising DSO suggests deteriorating credit quality of the customer base or, more seriously, fictitious revenue recording.

FURTHER INVESTIGATION: The analyst should request an ageing analysis of trade receivables (what proportion is more than 90/120 days overdue?), details of the company's revenue recognition policy and any changes thereto, and an explanation from management for the divergence between earnings and cash flow.

The analyst should also compare DSO to that of industry peers.

Section B — Multiple Choice Questions: Answer Key

Question	Answer
MCQ 1	B
MCQ 2	C
MCQ 3	C
MCQ 4	C
MCQ 5	B
MCQ 6	B

Section C — True / False Questions: Answers and Explanations

1. FALSE — FSA uses historical data as the empirical foundation for forward-looking judgments. The ultimate purpose of analysis is always to support decisions about the future.
2. TRUE — The notes disclose accounting policies, estimation assumptions, composition of aggregated line items, and contingent items that are essential for rigorous analysis.
3. FALSE — Identical economics reported under different accounting standards can produce materially different ratios. The Allianz/Volkswagen examples illustrate this clearly.
4. TRUE — As noted in the framework, the best analysts simultaneously pursue all four objectives, because each dimension informs and disciplines the others.
5. TRUE — A rising (and high) ratio of OCF to net income indicates that a growing share of reported earnings is backed by real cash, a key indicator of earnings quality.
6. FALSE — Common-size analysis removes the distortion of absolute size differences, but it does not resolve differences arising from accounting standard choices, business model variation, or industry-specific factors.

Section D — Real-World Interpretation Exercise

Exercise 4 — Integrated Analysis — Applying the FSA Framework

a) Key Ratios:

Gross Margin: FY2021: 30.0% | FY2022: 28.0% | FY2023: 26.0% [DECLINING]

EBIT Margin: FY2021: 14.9% | FY2022: 12.0% | FY2023: 8.9% [DECLINING SHARPLY]

OCF/NI: FY2021: 112.2% | FY2022: 94.1% | FY2023: 64.3% [DETERIORATING]

b) Organic revenue growth FY2023:

Reported revenue FY2023: €1,220m; Less acquired: -€120m = Organic: €1,100m

Organic growth: $(1,100 - 1,050) / 1,050 = +4.8\%$ (vs. reported +16.2%)

c) Four signals of concern:

SIGNAL 1 — Earnings quality deterioration: OCF/NI has fallen from 112% to 64%. Revenue is growing (partly inorganically), but cash generation is collapsing. This is a classic red flag for aggressive revenue recognition or receivables issues.

SIGNAL 2 — Rising receivables and likely DSO deterioration: Trade receivables have grown from €120m to €200m (+67%) while organic revenue grew only ~16% over the same period. DSO: FY21 = 46d; FY22 = 54d; FY23 = 60d. Combined with the changed revenue recognition policy, this warrants investigation.

SIGNAL 3 — Margin compression despite revenue growth: Gross and EBIT margins are declining consistently. Management calls this a 'year of investment', but the data suggests structural margin pressure — potentially a loss of competitive positioning or rising input costs that cannot be passed on. Competitor analysis is critical here.

SIGNAL 4 — Rapid leverage build-up: Net debt has more than doubled in two years (€220m to €520m) while EBIT has fallen from €142m to €109m. Net Debt/EBIT has risen from approximately 1.5x to nearly 4.8x. This is approaching levels that would concern credit rating agencies.

d) Perspective divergence:

EQUITY ANALYST: Would focus on the sustainability of the margin decline and the quality of earnings growth. Key question: is the revenue recognition policy change accelerating revenues that will not be repeated? Would seek detailed segment data, receivables ageing, and comparison of capex returns vs. industry peers.

CREDIT ANALYST: Would focus on the rapid leverage increase and the collapse of cash flow. Key question: can the company service €520m of net debt on €109m of EBIT and €45m of operating cash flow? Would seek the debt maturity schedule, covenant headroom details, and stress-test the business under a 20% revenue decline.

Chapter 2 — Reformulating the Income Statement: Core Operating Income, Non-Core Operating Income, and Unusual Item

Exercise 1 — Reformulating the Income Statement

(a) Reformulated Income Statement

Reformulated Income Statement — Manifattura SpA, 20X2	€ thousands
+ Net Revenues	350,000
– Cost of Goods Sold	(220,000)
– Selling Expenses	(35,000)
– General and Administrative Expenses	(20,000)
– Depreciation and Amortisation (operating)	(18,000)
(A) Core Operating Income	57,000
+ Dividends from listed equity stake	1,500
+ Rent revenues from Investment Properties	3,200
– Maintenance costs on Investment Properties	(800)
(B) Non-Core Operating Income	3,900
= Operating Income (A + B)	60,900
+ Interest Income (S-T marketable securities)	900
– Interest Expense (financial loans)	(7,500)
= Net Financial Expenses	(6,600)
= Income before Unusual Items and Taxes	54,300
+ Gain on Sale of Production Plant (unusual)	8,000
– Restructuring Charges (unusual)	(5,000)
– Impairment Loss on Intangibles (unusual)	(2,500)
± Unusual Items (net)	500
= Income before Taxes (EBT)	54,800
– Income Taxes (effective rate 30%)	(16,440)
= Net Income	38,360

(b) Core Operating Income = €57,000k; Non-Core Operating Income = €3,900k; Operating Income = €60,900k; Net Financial Expenses = €6,600k; Income before Unusual Items and Taxes = €54,300k; NOPAT = $60,900 \times 0.70 = €42,630k$.

(c) Dividends on the 4% listed stake and the rent/cost on Investment Properties are Non-Core Operating (income from financial assets and IP). Interest income and interest expense are Net Financial Expenses. The gain on the plant sale, the restructuring charge and the intangible impairment are Unusual Items (operating in nature, but non-recurring).

(d) The proposal is wrong for three reasons. First, the reported €38,360k includes net Unusual Items of €500k (≈€350k after tax) that are non-recurring and should not be projected. Second, it does not separate the core engine (€57,000k) from Non-Core income (dividends, IP), which should be projected on different assumptions. Third, even a correctly reformulated forecast must be benchmarked against the Cash Flow Statement: if recurring Core Operating Income is not converting into operating cash flow, the forecast is not credible.

Exercise 2 — The Net Financial Expenses Caution

(a) Net Financial Position = $5,000 - 2,000 = 3,000$; Net Financial Expenses = $250 - 60 = 190$; naïve ratio = $190 / 3,000 = 6.33\%$.

(b) Average cost of gross debt = $250 / 5,000 = 5.0\%$.

(c) After liquidation: debt = 3,000, assets = 0, NFP = 3,000; new Financial Expenses = $3,000 \times 5.0\% = 150$; Net Financial Expenses = 150; implied true cost = $150 / 3,000 = 5.0\%$.

(d) The naïve ratio (6.33%) exceeds both the cost of gross debt (5.0%) and the true cost of net debt (5.0%) because numerator and denominator are netted against amounts earning/costing different rates, and because the €60 of financial income mixes interest income (comparable to a cost) with dividends (not contractually due). This is the caution discussed in §1.5.

Exercise 3 — Net Income vs. Comprehensive Income (the Bond Example)

(a) and (b): see Table 2.5 in §3.3. Under historical cost, 20X0 Net Income = +5 (interest only), OCI = 0, Comprehensive Income = +5. Under FVTPL, Net Income = +12 (interest +5 and revaluation to fair value +7), OCI = 0, Comprehensive Income = +12. Under FVTOCI, Net Income = +5 (interest only), OCI = +7 (unrealised gain from the change in fair value), Comprehensive Income = +12. The split between Net Income and OCI differs, but in 20X0 the two fair-value choices both report a Comprehensive Income of +12.

(c) On the 20X1 sale at 109, for a bond classified as FVTOCI the €7 previously recognised in OCI is recycled into Net Income, so the total gain flows through profit or loss over the two periods — choices 1, 2 and 3 converge by the sale date. If instead the instrument were shares classified as FVTOCI, recycling into Net Income is not allowed (IFRS 9): the cumulative gain previously in OCI is transferred to another equity account on disposal, never passing through Net Income. This is precisely why the analyst must read the Statement of Changes in Equity (see §3.5) to capture all components of value creation.

Multiple-Choice Questions

MCQ 1. Correct Answer: B — Option B reproduces the definition in §1.2–1.3: Core Operating Income comes from sales related to the main business units only. Option A describes the total Operating Income (A+B). Option C mixes operating and financial. Option D is the bottom line, not COI.

MCQ 2. Correct Answer: C — Option C is one of the OCI items listed by IAS 1 (§3.2). Options A, B and D are examples cited in §3.3 of unrealised gains/losses recorded ABOVE Net Income, inside the regular Income Statement, not in OCI.

MCQ 3. Correct Answer: A — Option A is correct for a bond (§3.3): recycling into Net Income occurs on disposal. Section 3.3 stresses that for SHARES classified as FVTOCI recycling is NOT allowed — a key distinction. Options B, C and D misstate the recycling mechanics.

MCQ 4. Correct Answer: B — Option B reproduces Penman's argument quoted in §3.5. Option A is false (OCI items are audited). Option C is false (both IFRS and US GAAP have OCI). Option D is false (OCI can be positive or negative regardless of profitability).

MCQ 5. Correct Answer: C — Option C corresponds to the analytical warning in §4.4: an item occurring every year is no longer unusual in frequency, regardless of the company's label. Option A applies the label mechanically. Option B is arbitrary. Option D is wrong — restructuring is operating, not non-core.

MCQ 6. Correct Answer: B — Option B reproduces Penman's reasoning (§5.2 and the Abbott case): R&D is a core expense, and a reduction as a percentage of sales is a red flag of a discretionary variation that may inflate current operating income. Option A is wrong (R&D is operating). Option C is false. Option D overstates: a red flag warrants investigation, not an automatic fraud conclusion.

True / False Questions

T/F 1. TRUE. This is reformulation choice (c) in §1.2: Unusual Items — whether operating or financial in origin — are isolated on a single line just above Income before Taxes, so that the line 'Income before Unusual Items and Taxes' measures the persistent result.

T/F 2. FALSE. Section 3.3 stresses the opposite: many fair-value adjustments are recorded ABOVE Net Income, inside the regular Income Statement — for example FVTPL marketable securities, non-hedging/fair-value-hedge derivatives, Investment Properties under the fair-value option, revaluation deficits and impairments. Whether a fair-value change goes to Net Income or to OCI depends on the accounting rule for the specific asset class.

T/F 3. FALSE. Under IAS 21 the translation adjustment is accumulated in OCI, not Net Income (§3.2). Assets and liabilities of a foreign subsidiary are translated at the balance-sheet-date rate and revenues and expenses at the average rate; the difference is the translation adjustment, an OCI item.

T/F 4. TRUE. This is the substance of the Coca-Cola case (§2.4). The equity method is compliant; the analytical treatment considers proportional consolidation because the investment is economically part of the core business, with material consequences for operating scale, leverage and ROI.

Real-World Interpretation Exercise — Trend Analysis and the Discretionary Variation

(a) Sales growth decelerated from 11.1% (Year 2) to 5.0% (Year 3). At the same time, advertising/Sales fell from 6.0% to 4.0% and R&D/Sales fell from 9.0% to 8.0%. Both core discretionary expenses were cut as a percentage of sales precisely as growth slowed — the Abbott pattern.

(b) The cuts in advertising and R&D are category-3 discretionary variations: controlled by management in the short term and, here, coinciding with a slowdown in sales — a red flag that they may have been used to support reported earnings.

(c) Recurring advertising = $2,100 \times 6.0\% = \text{€}126\text{m}$ (actual €84m; €42m less). Recurring R&D = $2,100 \times 9.0\% = \text{€}189\text{m}$ (actual €168m; €21m less). Total discretionary saving = €63m. Recurring Operating Income = $168 - 63 = \text{€}105\text{m}$, which is below Year 2's €120m and Year 1's €126m. The reported Year-3 improvement (€120m → €168m, +40%) reverses into a decline once the discretionary cuts are normalised.

(d) The headline Operating Income growth is not driven by the core business; it is largely the product of cutting two core discretionary expenses as growth slowed. As an equity analyst I would not extrapolate the Year-3 result. The single question to management: are the reductions in advertising and R&D a permanent efficiency gain, or a temporary measure to support earnings — and, if temporary, what is the expected impact on future sales? (This mirrors the central question Penman asks in the Abbott case.)

Chapter 3 — Analysing the Operating Income: Three Reformulations of the Operating Cost Structure

Section A — Numerical Exercises

A.1 (a) Total manufacturing costs = 14,000 + 9,000 + 7,000 = 30,000. COGS = 30,000 + beginning inventory 2,000 – ending inventory 5,000 = 27,000. Gross Profit = 50,000 – 27,000 = 23,000. (b) Gross Profit Margin = 23,000 ÷ 50,000 = 46.0%. Core Operating Income = 23,000 – 8,000 – 6,000 – 4,000 = 5,000; Operating Margin = 5,000 ÷ 50,000 = 10.0%. (c) The gross margin depends on what the firm places inside COGS. If a competitor classifies, for instance, inbound freight or supervisory salaries below the gross line while this firm includes them in COGS, the two gross margins differ for reasons of classification, not efficiency.

Economic meaning: 46 cents of each revenue euro survive production to cover the selling, administrative and research spending that consumes 36 cents, leaving 10 cents of operating profit. *Common mistake:* treating COGS as production cost; it is the cost of goods *sold* — production cost adjusted for the inventory change — so the €3,000 inventory build reduced COGS below the €30,000 produced.

A.2 (a) and (b):

(€)	Firm Alpha	Firm Beta
Value of Production	200	200
– External purchases	(80)	(140)
= Added Value	120	60
– Cost of personnel	(70)	(25)
= EBITDA	50	35
– Depreciation & amortisation	(30)	(15)
= Core Operating Income	20	20
Added Value ÷ Value of Production	60.0%	30.0%
Personnel ÷ Added Value	58.3%	41.7%
D&A ÷ EBITDA	60.0%	42.9%
EBITDA margin (÷ Value of Production)	25.0%	17.5%

(c) Alpha is far more internalised (Added Value 60% versus 30%): it creates most of its value in-house and carries heavy internal costs. Consequently Alpha is more exposed to a wage round (personnel is 58.3% of its Added Value) and to the cost of its asset base. Beta is more externalised: it buys 70% of its value from suppliers and is therefore more exposed to a surge in raw-material prices, which would strike its large external-cost line directly and could not be diluted by internal value creation. Same operating income; opposite exposures. *Common mistake:* reading a higher Added Value as higher profitability (see C.1) — both firms earn 20.

A.3 (a) Revenue = $10,000 \times 40 = 400,000$. Variable costs = $10,000 \times 24 = 240,000$ (60% of sales). Contribution Margin = $400,000 - 240,000 = 160,000$ (unit CM = €16). Operating income = $160,000 - 120,000 = 40,000$. DOL = $160,000 \div 40,000 = 4.0$. Margin of Safety = $40,000 \div 160,000 = 25.0\%$. (b) Predicted $\% \Delta \text{OI} = 15\% \times 4.0 = 60\%$, so $\text{OI} \approx 64,000$. Verification: revenue 460,000, variable costs 276,000 (60%), Contribution Margin 184,000, fixed costs 120,000, operating income 64,000 — an increase of $(64,000 - 40,000)/40,000 = 60\%$, confirming the elasticity. (c) Break-even units = fixed costs €120,000 $\div$ unit contribution margin €16 = 7,500 units. Break-even sales = $7,500 \times €40 = 300,000$, or fixed costs 120,000 $\div$ contribution-margin ratio 40% = 300,000. Margin of Safety = $(400,000 - 300,000)/400,000 = 25\%$, identical to $\text{OI} \div \text{CM}$.

Economic meaning: each euro of sales contributes 40 cents toward fixed costs and then profit; the firm can lose a quarter of its revenue before operating profit vanishes. *Common mistake:* applying the DOL to a revenue change so large that fixed costs would in reality step up, or that pushes the firm outside the relevant range — the linear elasticity holds only locally.

A.4 (a) Raw-material consumption 12,000 is 37.5% of net revenues ($\div 32,000$), 32.4% of the value of production ($\div 37,000$), and 48.0% of total manufacturing cost ($\div 25,000$). (b) After the €2 price rise, revenue = $800 \times €42 = 33,600$ while every cost is unchanged. Raw materials $\div$ net revenues = $12,000 \div 33,600 = 35.7\%$ (down from 37.5%); raw materials $\div$ total manufacturing cost = $12,000 \div 25,000 = 48.0\%$ (unchanged). (c) A mechanical reader watching “raw materials as a percentage of sales” would applaud an apparent improvement in purchasing efficiency. Nothing of the kind occurred: not one euro of cost moved. The percentage fell only because the revenue base grew through pricing. Measured against total manufacturing cost — a base independent of selling price — the ratio is flat, correctly showing that the production cost structure did not change.

Common mistake: attributing a movement in a cost-to-sales percentage to cost behaviour when the cause lies entirely in the denominator.

Section B — Multiple-Choice Questions

B.1 — (b). If the gross margin rises but the operating margin does not, costs must have shifted out of COGS and into the lines below the gross profit, lifting the upper margin while leaving the lower one untouched. (a) is wrong because a genuine production improvement combined with proportional SG&A cuts would lift the operating margin too. (c) and (d) concern financing, which lies below the operating section and cannot move an operating margin.

B.2 — (b). A high Added-Value ratio means the firm creates value internally and carries heavy internal (personnel and asset) costs, so it is more exposed to wage and asset-cost shocks; a low ratio means reliance on external purchases and greater exposure to input-price increases. (a) is the classic trap — equal operating income means equal profitability here. (c) confuses the internal/external axis with the variable/fixed axis. (d) does not follow: the ratio says nothing about the level of revenue.

B.3 — (b). EBITDA is only an approximation of operating cash flow: it adds back non-cash D&A but ignores the cash absorbed by working-capital growth and the future cash needed to replace assets. (a) overstates the relationship. (c) is false — EBITDA is Added Value minus personnel, hence smaller than Added Value. (d) is false — lease accounting directly affects whether a cost appears as rent (above EBITDA) or as depreciation and interest (the motivation for EBITDAR/EBITDAaL).

B.4 — (b). The DOL is an elasticity equal to $CM \div OI$; as a firm earns more operating profit it sits further above break-even and its DOL falls. The lower number signals greater safety, not weakness. (a) inverts the meaning. (c) would change the contribution-margin ratio and is not implied. (d) would, if anything, raise the DOL, not lower it.

B.5 — (b). A cost-to-sales ratio that falls while the same cost measured against an internal base (manufacturing cost) is unchanged points to a growing denominator — higher prices or volumes — not to genuine cost improvement. (a) would also reduce the manufacturing-cost-based ratio. (c) and (d) concern other cost categories or accounting methods and would not produce this specific pattern.

B.6 — (b). EBITDAR adds back rentals as well as depreciation so that firms which lease their productive assets can be compared on operating terms with firms that own them. (a) is the opposite of its purpose — it neutralises a financing-style difference rather than comparing financing. (c) and (d) are unrelated.

Section C — True / False Questions

C.1 — False. The critical word is “always.” Added Value measures the degree of internalisation, not profitability; two firms can share an operating income while one has a far higher Added Value (see A.2). A higher Added Value simply means more value is created in-house and more internal cost is carried.

C.2 — False. The flaw is the word “equal.” EBITDA adds back D&A and so removes one non-cash item, but it ignores changes in working capital, other non-cash items, and the eventual cash cost of replacing assets. EBITDA approximates operating cash flow only under assumptions that seldom hold; the two are not equal.

C.3 — True. Holding operating income fixed, a higher Contribution Margin must be matched by higher fixed costs (since $OI = CM - \text{fixed costs}$). A larger fixed-cost share is precisely what makes a cost structure rigid, so profit will swing more violently with volume.

C.4 — False. The DOL is an elasticity, not a profitability measure. A fall usually means the firm has moved away from break-even and is now more profitable and less profit-elastic — generally good news. Reading a falling DOL as deteriorating profitability is a standard error.

C.5 — False. The word “directly” is the problem. A manufacturer’s COGS contains materials, labour and overhead built through cost accounting; a retailer’s contains only the purchase cost of merchandise. The two gross margins answer different questions and are comparable only after their COGS definitions are reconciled.

Section D — Real-World Interpretation Exercise

D.1 (a) Year 2: Contribution Margin = $2,988.1 - 893.9 - 336.3 - 459.1 = 1,298.8$ (43.5%); operating profit 402.1 (13.5%); DOL = $1,298.8 \div 402.1 \approx 3.2$; Margin of Safety = $402.1 \div 1,298.8 \approx 30.9\%$. Year 1: Contribution Margin = $2,942.0 - 1,257.1 - 286.6 - 443.4 = 954.9$ (32.5%); operating profit 144.2 (4.9%); DOL = $954.9 \div 144.2 \approx 6.6$; Margin of Safety $\approx 15.1\%$. (b) Good news: the DOL fell because operating profit nearly tripled, moving the carrier far from break-even — the Margin of Safety doubled from 15.1% to 30.9%. A lower elasticity here reflects greater, safer profit. (c) Revenue was essentially flat (+1.6%), so almost none of the margin improvement came from selling more. The dominant driver was the fall in fuel from 42.7% to 29.9% of revenue — a price effect, not a volume effect. This shows that the contribution-margin reformulation, which presumes costs flex with a single volume, is unreliable for this firm: one of its largest “variable” costs is driven by a volatile external price independent of volume. (d) Strip fuel out — the ex-fuel contribution margin (73.4% versus 75.2%) is a far steadier series revealing the underlying operating economics — and replace the single DOL with multi-driver sensitivity analysis (varying flights, load factor, passengers, and fuel price separately) and Monte Carlo simulation of their joint distribution. An equity analyst would value the carrier on fuel-hedged, ex-fuel margins; a credit analyst would stress-test debt service against simultaneous fuel-price and demand shocks.

Common mistake: declaring the firm riskier because its DOL fell, and crediting management with an operating improvement that was in fact a fall in the world price of jet fuel.

Chapter 4 — Reformulating the Balance Sheet: From the “Liquidity” Format to the “Managerial” Format

Numerical Exercises

Exercise 1 — Solution. (a) Short-Term Assets = $600 + 1,400 + 1,000 + 400 = 3,400$. Short-Term Liabilities = $1,100 + 200 + 300 + 900 = 2,500$. Long-Term Assets = $6,600$. Long-Term Liabilities = $3,000$; Net Equity = $4,500$.

(b) Treasury Margin = $(600 + 1,400 + 400) - 2,500 = -100$ (< 0 , fails). Quick Ratio = $(600 + 1,400 + 400) \div 2,500 = 2,400 \div 2,500 = 0.96$ (< 1 , fails narrowly). NWC = $3,400 - 2,500 = 900$ (> 0 , passes). Current Ratio = $3,400 \div 2,500 = 1.36$ (below ≈ 2).

(c) Total Liabilities = $2,500 + 3,000 = 5,500$; Debt-to-Equity = $5,500 \div 4,500 = 1.22$ (≈ 1 , acceptable). Coverage of Long-Term Assets = $4,500 \div 6,600 = 0.68$ (< 1 , fails). Structure Margin = $4,500 - 6,600 = -2,100$ (< 0 , fails).

(d) The firm passes the crude NWC and debt-to-equity tests but fails the more demanding ones: the negative Treasury Margin and sub-one Quick Ratio show that cash and receivables alone do not cover short-term liabilities (inventory must be sold first), and the negative Structure Margin shows that long-term assets are partly financed by long-term debt rather than by permanent capital alone — a structural fragility the headline Current Ratio conceals. Consistent with Section 4.3.6, no conclusion is final until the cash-flow statement confirms the firm's ability to service its debt.

Exercise 2 — Solution. (a) DSO = $900 \div (3,600 \div 360) = 900 \div 10 = 90$ days. DIO = $600 \div (2,400 \div 360) = 600 \div 6.667 = 90$ days. DPO = $600 \div (1,800 \div 360) = 600 \div 5 = 120$ days. CCC = $90 + 90 - 120 = 60$ days.

(b) NOWC = $A/R + \text{Inventory} - A/P = 900 + 600 - 600 = 900$. NOWC / Net Revenues = $900 \div 3,600 = 25.0\%$.

(c) Because the ratio is invariant to the sales level when the days-in and cost structure are held constant, new NOWC $\approx 25.0\% \times (3,600 \times 1.25) = 25.0\% \times 4,500 = 1,125$. Additional working-capital investment = $1,125 - 900 = 225$ — cash the firm must finance before the profit on the extra sales is collected.

(d) The CCC of +60 days sums non-comparable quantities (one day of receivables, inventory and payables each correspond to different euro amounts), so it cannot be turned into a financing figure directly; the NOWC/Revenues ratio, by contrast, is a stable, predictive percentage that converts a revenue forecast straight into a working-capital and financing forecast — the reasoning endorsed in Section 4.3.5.

Exercise 3 — Solution. (a) Current Operating Assets = $A/R\ 2,000 + \text{Inventory}\ 1,500 = 3,500$. Operating Liabilities = $A/P\ 1,300 + \text{operating provisions}\ 400 = 1,700$. NOWC = $3,500 - 1,700 =$

1,800. (Tax payables of 300 are treated here as a separate non-operating item, per the options of Section 4.4.3.)

(b) Net Operating Invested Capital = NOWC 1,800 + Non-Current Operating Assets 5,000 = 6,800.

(c) Net Financial Position = financial liabilities – liquid financial assets = (1,200 + 500 + 2,800) – (cash 200 + marketable securities 800) = 4,500 – 1,000 = 3,500.

(d) Operating side: NOIC 6,800 + non-operating long-term investments 1,000 = 7,800, less tax payables 300 = 7,500 of net invested capital to be financed. Financing side: NFP 3,500 + Net Equity 4,000 = 7,500. The two sides balance. Financial Debt-to-Equity (using NFP) = 3,500 ÷ 4,000 = 0.875. As Section 4.4.3 notes, a different but equally acceptable format would fold the tax payable into net operating assets; the balancing identity is preserved either way.

Multiple-Choice Questions

MCQ 1. Correct answer: B — Option B reproduces the managerial-format definition. Option A is the Net Working Capital of the liquidity format (a different concept). Option C is net assets/equity. Option D mixes operating assets with financial liabilities, violating the operating/non-operating sorting.

MCQ 2. Correct answer: C — Option C is this chapter's warning: a high Current Ratio is a question, not a verdict, and each swollen asset or thin liability must be explained by a business reason or treated as a possible accounting issue. Options A and B both read a single number mechanically; option D is not a remedy.

MCQ 3. Correct answer: B — Option B reproduces the NFP construction: liquid financial assets are netted into financial debt and simultaneously removed from the surplus-asset side, preserving the balance. Netting all financial assets (short- and long-term) instead yields Net Financial Obligations. Options A, C and D describe unrelated operations.

MCQ 4. Correct answer: B — Option B is the decisive objection: the three days-in components are measured against different income-statement flows, so summing them is dimensionally unsound. The ratios use closing balances (so A is false); the CCC can be negative (so D is false); taxes are not the issue (so C is false).

MCQ 5. Correct answer: B — Option B reflects the arithmetic in the simulation: reducing the numerator and denominator of a ratio by an equal amount raises the ratio when it exceeds 1 (here 1.08 → 1.10). If the ratio were below 1 the same action would lower it; the Quick Ratio identity in C is irrelevant.

MCQ 6. Correct answer: C — Option C states this chapter's rationale: the managerial format underpins ROI/ROE, leverage analysis and the change-in-NOWC that drives cash-flow

reformulation and valuation. The managerial format does not produce a Current Ratio (A); IAS 1 does not mandate it (B); it requires more deliberate classification, not less (D).

True / False Questions

T/F 1. FALSE. “Short-term” is a maturity criterion (within twelve months); “current operating” is a nature criterion (belonging to the operating cycle) and may include items realised beyond twelve months. This chapter stresses that “current” $\neq$ “short-term”, and neither coincides with the IAS 1 definitions.

T/F 2. FALSE. This chapter's three simulations show a firm with a CCC of –30 days that still has a positive NOWC; only in extreme cases (a strongly negative cycle) does NOWC itself turn negative.

T/F 3. TRUE. This invariance is exactly what the simulations demonstrate and what gives the ratio its predictive value: it stays at 25%, 10% or –9.44% whether revenues are 100 or 120.

T/F 4. TRUE. This is the window-dressing limitation: bringing forward payments, drawing long-term loans to hold as cash and bridging operations across the year-end all lift the ratios without changing the underlying business.

T/F 5. FALSE. The items-versus-accounts limitation shows that the structural, permanent mass of these accounts cannot in practice be split; analysts treat the full balance as short-term and study its change through trend analysis and the reformulated cash-flow statement. This is one of the reasons they prefer the managerial format.

Real-World Interpretation Exercise

Exercise — One Current Ratio, two structural stories. Solution. (a) Alfa Quick Ratio = $(100 + 700) \div 2,000 = 0.40$; Beta = $(900 + 400 + 600) \div 2,000 = 0.95$. Identical Current Ratios conceal very different short-term liquidity: Alfa's working capital is locked in inventory (1,200), which must be sold and collected before it covers payables; Beta's is in cash and securities, immediately available. The Quick Ratio exposes what the Current Ratio hides.

(b) Beta. The €700m long-term loan drawn in late December and held as cash is the textbook action b) of the window-dressing simulation (Section 4.6.1): it inflates short-term assets and the current and quick ratios while touching only the balance sheet, and it can be repaid or redeployed in January. Alfa shows no such pattern.

(c) Alfa's NOWC/Revenues, stable at 22%, indicates a capital-intensive working-capital model whose financing need scales predictably with sales — high but understood. Beta's rise from 8% to 14% is the warning sign: its operating business is absorbing progressively more working capital per euro of sales, which (independently of the window dressing) means future growth will consume more cash than the headline ratios suggest.

(d) As an equity analyst I would not read Beta's year-end liquidity as structural and would model its rising working-capital intensity into the cash-flow forecast; as a credit analyst I would discount the engineered December liquidity and test debt service against operating cash flow, per Section 4.3.6. The single question to Beta's management: is the late-December borrowing a permanent change in financing or a year-end bridging operation, and what explains the deterioration in NOWC/Revenues? The single question to Alfa's management: given the stable 22% intensity, what incremental financing is planned to fund the working capital of the next growth phase?

Chapter 5 — The Managerial Balance Sheet in Depth: Reformulation Options, Net Financial Position, and Structural Ratios

Numerical Exercises

Exercise 1 — Three measures of debt from one balance sheet.

Solution. Total financial debts = $800 + 400 + 2,600 = 3,800$. (a) Net Short-Term Financial Debt = short-term debts – cash – short-term financial assets = $(800 + 400) - 300 - 500 = 400$. (b) Net Financial Position = total debts – cash – short-term financial assets = $3,800 - 300 - 500 = 3,000$. (c) Net Financial Obligations = total debts – all financial assets = $3,800 - (300 + 500 + 1,200) = 1,800$.

(d) Net Short-Term Financial Debt (400) measures short-term financial exposure. The Net Financial Position (3,000) nets only liquid financial assets and is the figure used in covenants (e.g. NFP/EBITDA) and to estimate the cost of debt; the long-term equity stakes of 1,200 remain as surplus assets. The Net Financial Obligations (1,800) nets all financial assets, including the long-term stakes, and is used in the Modigliani–Miller decomposition of ROE — so the surplus-asset category disappears. The three differ precisely by which financial assets are netted against debt.

Exercise 2 — Working cash and the Operating Invested Capital.

Solution. (a) Working cash = $0.5\% \times 20,000 = 100$; surplus cash = $700 - 100 = 600$. (b) With working cash operating: Current operating assets = $2,300 + 100 = 2,400$; NOWC = $2,400 - 2,600 = -200$ (negative, financed by suppliers/customers); Net Operating Invested Capital = $-200 + 5,000 = 4,800$; surplus cash of 600 sits among non-operating assets.

(c) All cash surplus: Current operating assets = 2,300; NOWC = $2,300 - 2,600 = -300$; Net Operating Invested Capital = $-300 + 5,000 = 4,700$; surplus cash of 700 among non-operating assets.

(d) The working-cash treatment raises Operating Invested Capital by 100 (from 4,700 to 4,800), which slightly lowers any return-on-operating-investment ratio computed on it; the alternative treatment leaves the operating denominator smaller and instead reduces net financial debt. For a retailer the working-cash treatment is the more faithful, because the till cash is genuinely needed to run the operating cycle. The choice must be applied consistently across comparables (Section 5.3).

Exercise 3 — The three polar structures and the financing identity.

Solution. (a) Firm X: NFO = $6,000 - 1,000 = 5,000$ (obligations). Firm Y: $500 - 2,500 = -2,000$, i.e. Net Financial Assets of 2,000. Firm Z: $400 - 5,400 = -5,000$, i.e. Net Financial Assets of 5,000.

(b) Firm X: $5,000 \text{ (NFO)} + 5,000 \text{ (NE)} = 10,000 = \text{NOA}$. ✓ Firm Y: $\text{NOA } 4,000 = \text{NE } 6,000 - \text{NFA } 2,000$; i.e. $\text{NOA} + \text{NFA} = \text{NE}$ ($4,000 + 2,000 = 6,000$). ✓ Firm Z: $\text{NFA } 5,000 = \text{NE } 2,000 - \text{NOA } (-3,000) = 2,000 + 3,000$. ✓

(c) Firm X is the orthodox leveraged operating business (the General Mills case). Firm Y has a positive operating investment plus net financial assets, all funded by equity — a self-funded net lender (the Nike case). Firm Z has a negative net operating investment that, with equity, finances a large pile of net financial assets — the cash-generating model with negative NOWC (the Dell case).

Exercise 4 — Goodwill adjustment to leverage and ROE.

Solution. (a) Reported: financial leverage = $9,000 \div 4,000 = 2.25$; total liabilities = $9,000 - 4,000 = 5,000$, so liabilities/equity = $5,000 \div 4,000 = 1.25$; ROE = $600 \div 4,000 = 15.0\%$.

(b) Adjusted assets = $9,000 - 2,800 = 6,200$; adjusted equity = $4,000 - 2,800 = 1,200$. Adjusted financial leverage = $6,200 \div 1,200 = 5.17$; liabilities/equity (adjusted) = $(6,200 - 1,200) \div 1,200 = 5,000 \div 1,200 = 4.17$; adjusted ROE = $600 \div 1,200 = 50.0\%$.

(c) Goodwill of 2,800 is 70% of reported equity; removing it transforms a conservatively geared firm (leverage 2.25, liabilities/equity 1.25) into a highly geared one on tangible capital (5.17 and 4.17), and more than triples the apparent ROE (from 15.0% to 50.0%, $\times 3.3$). As in the Disney case, the reported equity is largely an unamortised purchase premium exposed to impairment; the analyst should view the firm both ways (Section 5.8).

Multiple-Choice Questions

MCQ 1. Correct answer: B — Option B is the definition adopted in this chapter: working cash supports the purchase–conversion–sale cycle and is an operating asset, often estimated at about 0.5% of revenues for retailers; the surplus is non-operating. Option A ignores the split; C and D misdescribe it.

MCQ 2. Correct answer: C — Option C: Net Financial Obligations subtract total financial assets, so the non-operating-asset category disappears; it is used in the Modigliani–Miller ROE decomposition. The Net Financial Position nets only cash and short-term financial assets; the short-term measure nets only against short-term debt.

MCQ 3. Correct answer: B — Option B: “current = re-current (recurring)” — operating liabilities arise from the recurring operating cycle and can be short- or long-term. Option A confuses current with short-term; IAS 1 uses a hybrid definition (not “in all cases”), so C is wrong; D describes financial netting, not operating liabilities.

MCQ 4. Correct answer: C — Option C: a negative NOWC is the recurring net investment turning negative because the operating model is self-financing — the operating-leverage / Dell case. It is not a sign of insolvency or error, and it does not require zero inventory.

MCQ 5. Correct answer: B — Option B: if minority equity is treated as financial debt, the dividends paid to minorities are the cost of that financing and belong in financial expenses, and the reformulated bottom line must be the parent's net income. The other options are unrelated.

MCQ 6. Correct answer: C — Option C: removing goodwill from both assets and equity shrinks the (smaller) equity base proportionally more, so leverage ($\text{assets} \div \text{equity}$) and ROE ($\text{income} \div \text{equity}$) rise sharply — Disney's leverage $2.32 \rightarrow 6.51$ and ROE $21.7\% \rightarrow 90.8\%$. It does not touch the financial-asset/obligation distinction.

True / False Questions

T/F 1. FALSE. They differ by which financial assets are netted: the NFP subtracts only cash and short-term financial assets (long-term financial investments remain as surplus), while the NFO subtracts all financial assets (so non-operating assets disappear). They serve different uses — covenants/cost of debt versus the ROE decomposition.

T/F 2. TRUE. Operating liabilities are all “current” in the recurring sense and are deducted in full to form NOWC; they are not divided by maturity because the analyst sets them against the operating assets they spontaneously finance.

T/F 3. FALSE. IAS 1 expressly states that an entity shall not classify deferred tax assets (liabilities) as current; this is why the tax position is treated separately or folded whole into the operating side.

T/F 4. FALSE. The preferred treatment is the opposite: place the net held-for-sale amount among the non-operating (surplus) assets so that it does NOT inflate operating invested capital; if the net is negative, show it as an “other liabilities” category.

T/F 5. TRUE. The U.S. empirical study (876 firms) found average equity rising about 2% on reclassification, but over 25% for a tenth of firms, with comparable effects on income and leverage — so where minorities are large the choice is material and must rest on their economic role.

Real-World Interpretation Exercise

Exercise — One identity, three businesses, and a goodwill twist.

Solution. (a) Manifattura: $\text{NFO} = 5,000 - 500 = 4,500$; total equity (parent + minority) = 3,500; $4,500 + 3,500 = 8,000 = \text{NOA}$. ✓ Piattaforma: $\text{NFO} = 300 - 4,000 = -3,700$ (Net Financial Assets 3,700); $\text{NOA} - 2,000 = \text{equity } 1,700 - \text{NFA } 3,700$, i.e. $\text{NFA} = \text{NE} - \text{NOA} = 1,700 + 2,000 = 3,700$. ✓ Marchio: $\text{NFO} = 2,000 - 1,000 = 1,000$; $1,000 + 5,000 = 6,000 = \text{NOA}$. ✓

(b) Manifattura is the orthodox leveraged operating business (General Mills type). Piattaforma has a negative operating investment funding net financial assets — the Dell type, a self-

financing cash generator. Marchio is a moderately leveraged operating business whose equity is dominated by goodwill.

(c) Manifattura, minority as financial debt: financial debt-to-(parent)-equity = (financial debt + minority equity) ÷ parent equity = $(5,000 + 500) \div 3,000 = 1.83$, decomposing into financial leverage $5,000 \div 3,000 = 1.67$ plus shareholding leverage $500 \div 3,000 = 0.17$ (components $1.67 + 0.17 = 1.84$ differ from 1.83 by rounding). Coherence requires that minority dividends be reclassified as financial expenses and that the reformulated bottom line be the parent's net income (Section 5.7). Because the minorities are described as a dividend-paying financial partner, the debt treatment is the appropriate one.

(d) Marchio reported leverage on invested capital (invested capital ÷ equity): invested capital (assets net of operating liabilities) = NOA + financial assets = $6,000 + 1,000 = 7,000$; equity 5,000; leverage = 1.40. Tangible basis: adjusted invested capital = $7,000 - 3,500 = 3,500$; adjusted equity = $5,000 - 3,500 = 1,500$; tangible leverage = $3,500 \div 1,500 = 2.33$. The firm looks conservatively financed on reported figures but is materially more leveraged on tangible capital, and its equity is 70% goodwill exposed to impairment — the Disney lesson of Section 5.8. As an analyst I would monitor impairment indicators and read leverage on the tangible basis.

Chapter 6 — Return on Investment: Measuring the Profitability of Operations

A. Numerical Exercises

Exercise 1 — ROS, Asset Turnover and the two profitability engines

(a) $\text{ROI(A)} = 120 / 400 = 30.0\%$; $\text{ROI(B)} = 120 / 1,000 = 12.0\%$. Identical absolute profit and identical sales, yet A is more than twice as profitable on its capital.

(b) ROS is identical: $120 / 1,200 = 10.0\%$ for both. The difference is entirely in turnover. Asset Turnover(A) = $1,200 / 400 = 3.0\times$; Asset Turnover(B) = $1,200 / 1,000 = 1.2\times$. Check: $A = 10\% \times 3.0 = 30\%$; $B = 10\% \times 1.2 = 12\%$.

(c) Both earn the same margin, so the gap is a pure velocity gap. Firm A works a light capital base hard (3 turns) — an asset-light, high-velocity model. Firm B carries 2.5 times the capital for the same sales — a capital-intensive model whose extra assets are not (yet) generating proportional sales, suggesting over-investment, idle capacity, or a heavier fixed-asset/working-capital base. The decomposition tells the analyst to investigate B's balance sheet (fixed assets and NOWC), not its income statement.

Common mistake: concluding from equal operating income that the firms are equally profitable. ROI exists precisely to expose the capital difference the income statement hides.

Exercise 2 — Gross vs. net ROI and the operating-liability-leverage multiplier

(a) Net core invested capital = $500 - 200 = 300$. Gross core ROI = $90 / 500 = 18.0\%$. Net core ROI (RONA) = $90 / 300 = 30.0\%$.

(b) Core ROS = $90 / 600 = 15.0\%$. Gross turnover = $600 / 500 = 1.20$. Multiplier = gross / net capital = $500 / 300 = 1.667$. Check: $15.0\% \times 1.20 \times 1.667 = 30.0\%$. ✓

(c) Of the 30% RONA, the operating-asset engine ($15\% \times 1.20 = 18\%$) is genuine operating productivity; the remaining lift to 30% is contributed by the 1.667 multiplier — i.e. by the €200 of free supplier/accrual financing. The gross ROI of 18% is the honest measure of asset productivity and is independent of how the operating cycle is financed; the 30% RONA is real but partly a financing phenomenon.

Common mistake: reading the 30% as if it were all operating efficiency. Always isolate the multiplier before praising a high RONA.

Exercise 3 — CAPEX intensity and productive-capacity replacement

(a) CAPEX-to-Sales: Y1 6.0%, Y2 5.0%, Y3 8.8%. Productive Capacity Replacement (CAPEX/D&A): Y1 1.20, Y2 1.00, Y3 1.83.

(b) Year 1: replacement 1.20 — slightly above maintenance, modest development. Year 2: replacement 1.00 — pure maintenance, investment exactly replaces consumed capacity. Year 3: replacement 1.83 and CAPEX-to-Sales jumping to 8.8% — clear development investment well beyond consumption.

(c) The Year 3 surge signals the firm is building new capacity; if the investment is sound, future sales growth should follow as the new capacity comes on stream. The analyst would corroborate this against the industry-average CAPEX intensity and revisit it in the cash flow statement.

Common mistake: reading a single year's ratio in isolation. The maintenance/development distinction only emerges from the replacement ratio relative to 1 and from the trend against peers.

B. Multiple-Choice Questions

MCQ 1

Correct: B.

B is correct: net income is already reduced by interest and taxes, so dividing it by total assets mixes financing and tax effects with asset productivity — it measures neither operating nor shareholder return cleanly.

A is wrong — that role belongs to ROI (operating income / invested capital). C is wrong — the gap between ROA and ROI comes from the numerator (net vs. operating income) and denominator (total vs. operating capital), not from inventory. D is wrong — there is no general ordering; a highly taxed, heavily indebted firm will show ROA well below its ROI, but the relationship is not fixed.

MCQ 2

Correct: B.

B is correct: since $ROI = ROS \times \text{Asset Turnover}$ and ROS is unchanged, the rise must come from the turnover factor — either genuinely more sales per euro of capital, or a shrinking denominator.

A is wrong — unchanged ROS rules out a margin effect. C and D are wrong — ROI is an operating measure built on operating income and operating capital; taxes and interest do not enter it (they affect ROE and ROA, not ROI).

MCQ 3

Correct: B.

B is correct: the multiplier equals $1 / (1 - \text{Op.Liabilities/gross Op.Capital})$, so it exceeds 1 precisely when operating liabilities (trade payables, accruals) fund part of the operating assets.

A is wrong — financial debt affects ROE leverage, not the operating-liability multiplier, which concerns only operating liabilities. C is meaningless — ROS and turnover are not comparable in that way. D is wrong — the timing convention does not create the multiplier; operating liabilities do.

MCQ 4

Correct: B.

B is correct: when CAPEX falls short of depreciation, the firm is consuming productive capacity faster than it replaces it — under-investment that flatters current free cash flow at the expense of future capacity.

A is wrong — growth investment shows a ratio well above 1. C is wrong — the ratio carries clear economic meaning about future capacity. D is wrong — the ratio concerns investment, not margins.

MCQ 5

Correct: C.

C is correct: ending capital is the best proxy for the base the firm will carry into next year, so a return computed on it is the most prudent forecast of next-year profitability — and it requires no extra prior-year data for trend analysis.

A overstates performance (income earned partly after the new assets were added, divided by the small opening base). B is the coherent compromise but, with a large mid-year jump, a simple average misrepresents the capital actually available. D is wrong — with a growing base the three measures diverge materially, as the 10%/6.67%/5% example shows.

C. True / False Questions

T/F 1

TRUE.

Gross ROI uses gross operating assets in the denominator and ignores operating liabilities, so it is unaffected by the financing of the operating cycle — as the year-t/year-t+1 example shows, it does not change when the firm takes on supplier financing. Net ROI nets off operating liabilities and therefore rises with the operating-liability-leverage multiplier. The critical part is the contrast: it is the netting of operating liabilities that makes RONA financing-dependent.

T/F 2

FALSE.

The critical word is “always”. Net ROI can rise purely because the operating-liability-leverage multiplier grew (more free supplier/accrual financing) or because the asset base shrank through depreciation or impairment — neither of which is an operating-efficiency gain. Only decomposing the change into ROS, turnover and the multiplier reveals whether the improvement is genuine.

T/F 3

TRUE.

This is the general rule (see Robinson et al., International Financial Statement Analysis, CFA Institute, Wiley): a flow numerator spans the period while a stock denominator is a point-in-time figure, so averaging the denominator restores coherence. The qualification is that if a large change occurred at a known date, an interim-period average is better than a simple two-point average.

T/F 4

FALSE.

The name is unimportant; what matters is stating precisely what is in the numerator and denominator (core vs. non-core, before vs. after tax, operating vs. total, gross vs. net) and applying the same definition consistently. Two analysts using the same label but different definitions will get different, incomparable numbers.

T/F 5

FALSE.

The critical word is “always”. Because net fixed assets fall with depreciation and impairment ($\Delta = +\text{Acquisitions} - \text{Disposals} - \text{Depreciation} - \text{Impairment}$), the ratio can rise simply because the denominator shrank, not because sales per unit of capacity improved. Fixed-asset ratios are also unstable since assets are bought in large, infrequent lumps; the trend must be read with depreciation policy in mind.

D. Real-World Interpretation Exercise

Exercise — Diagnosing a “more efficient” company

Step 1 — compute the returns. Net operating capital: $Y1 = 1,250 - 250 = 1,000$; $Y2 = 1,240 - 440 = 800$. RONA: $Y1 = 200/1,000 = 20.0\%$; $Y2 = 205/800 = 25.6\%$. RONA indeed rose strongly, as claimed. Gross ROI: $Y1 = 200/1,250 = 16.0\%$; $Y2 = 205/1,240 = 16.5\%$ — essentially flat.

Step 2 — decompose. ROS: $10.0\% \rightarrow 10.0\%$ (flat). Gross turnover: $1.60 \rightarrow 1.65$ (marginal). Operating-liability-leverage multiplier: $1,250/1,000 = 1.25 \rightarrow 1,240/800 = 1.55$. The entire RONA

improvement (20.0% → 25.6%) comes from the multiplier rising from 1.25 to 1.55 — i.e. from operating liabilities jumping from 250 to 440. Margin and asset productivity barely moved.

Step 3 — read the investment signals. Productive Capacity Replacement fell from $100/110 = 0.91$ to $70/130 = 0.54$ — already below 1 and now sharply so. CAPEX-to-Sales fell from 5.0% to 3.4%. The firm is under-investing relative to the capacity it consumes.

Equity-analyst conclusion: the headline "capital efficiency" gain is illusory — it is stretched payables, not operating improvement. Worse, the firm is simultaneously under-investing (replacement 0.54), harvesting its asset base to flatter current returns and cash flow. The clean gross ROI, flat at ~16%, is the honest picture. This is not value creation through operations.

Credit-analyst conclusion: two warning signs at once. The return now depends heavily on €440m of supplier financing that could be withdrawn if terms tighten, and the firm is deferring maintenance CAPEX, storing up a future spending obligation that will compete with debt service. The apparent strengthening masks rising operating and liquidity fragility.

Common mistake: accepting the RONA improvement at face value. Decomposing it — and checking gross ROI and the replacement ratio — reverses the conclusion entirely.

Chapter 7 — Return on Equity and Its Decompositions: DuPont, Modigliani-Miller, Penman, and the Bridge to EVA

Exercise 7.1 (Numerical) — Full basic and Penman decomposition

(a) Income before Unusual Items and Taxes = $200 + 10 - 50 = 160$; Income before Taxes = $160 - 10 = 150$; Taxes (20%) = -30 ; Comprehensive Income = 120. After-tax statement: NOPAT = $200 \times 0.80 = 160$; Net Financial Expenses after taxes = $-40 \times 0.80 = -32$; Unusual Items after taxes = $-10 \times 0.80 = -8$; check: $160 - 32 - 8 = 120$ ✓. Managerial balance sheet: NOIC = $1,500 - 500 = 1,000$; Net Financial Debts = $750 - 250 = 500$; Net Equity = 500; check: $1,000 = 500 + 500$ ✓.

(b) **Basic formula.** $OI/NOIC = 200/1,000 = 20\%$; cost of net debt = $(50-10)/500 = 40/500 = 8\%$; leverage = $500/500 = 1$; tax & unusual ratio = $120/160 = 0.75$.

$$ROE = [20\% + (20\% - 8\%) \times 1] \times 0.75 = 32\% \times 0.75 = 24.00\%$$

Check: $120/500 = 24.00\%$ ✓.

(c) **Penman formula.** $RNOA = 160/1,000 = 16\%$; after-tax cost of net debt = $32/500 = 6.4\%$; unusual ratio = $120/(160-32) = 120/128 = 0.9375$.

$$ROE = [16\% + (16\% - 6.4\%) \times 1] \times 0.9375 = 25.6\% \times 0.9375 = 24.00\% \checkmark$$

RNOA decomposition: implicit interests = $500 \times 3\% = 15$; ROOA = $(160 + 15)/1,500 = 175/1,500 = 11.67\%$; operating liability leverage = $500/1,000 = 0.50$.

$$RNOA = 11.67\% + (11.67\% - 3.00\%) \times 0.50 = 11.67\% + 4.33\% = 16.00\% \checkmark$$

Economic meaning: the firm earns 11.67% on its gross operating assets, borrows implicitly from suppliers at 3% (operating spread 8.67%, multiplier 0.50 → +4.33 points), and explicitly from lenders at 6.4% after tax (financial spread 9.6%, multiplier 1 → +9.6 points); unusual items shave 6.25% off the result. Common mistake: using the gross 750 of financial liabilities (instead of the net 500) in the leverage term after having netted the expenses — numerator and denominator must be netted consistently.

Exercise 7.2 (Numerical) — The leverage effect and the negative spread

(a) $ROE = [12\% + (12\% - 5\%) \times 2] \times 0.70 = [12\% + 14\%] \times 0.70 = 18.20\%$.

(b) Spread = $4\% - 7\% = -3\%$. $ROE = [4\% + (-3\%) \times 2] \times 0.70 = (-2\%) \times 0.70 = -1.40\%$. The same multiplier that added 14 points in (a) now subtracts 6.

(c) Solve $[4\% - 3\% \times L] \times 0.70 = -10.5\% \rightarrow 4\% - 3\%L = -15\% \rightarrow L = 19/3 \approx 6.33$. Interpretation: with a -3% spread, a leverage of 6.33 wipes out more than ten points of equity return. Common

mistake: forgetting that the tax ratio also scales losses (a frequent slip is to apply 0.70 only to positive results).

Exercise 7.3 (Numerical) — DCF = EVA on a project

	Year 0	Year 1	Year 2	Year 3
Operating FCF (NOPAT + Depr. – CAPEX)	–300.0	145.0	145.0	145.0
Discount factor (10%)	1.0000	0.9091	0.8264	0.7513
Discounted FCF	–300.0	131.8	119.8	108.9
NPV (DCF)	60.6			
Beginning Invested Capital		300.0	200.0	100.0
EVA = NOPAT – 10% × Capital		15.0	25.0	35.0
Discounted EVA		13.6	20.7	26.3
NPV (EVA)	60.6			

Economic meaning: capital declines by exactly the depreciation each year (no CAPEX after time 0, no NOWC change), which is the consistency condition of §7.9 under which DCF and EVA must coincide. Common mistake: including the –300 investment in the EVA stream — the capital charge (10% × beginning capital) already remunerates it.

Multiple-Choice Questions

MCQ 1 — Correct: (c). (c) shareholders own both income streams, including the dirty-surplus items routed through OCI (Chapter 2). (a) ignores OCI and understates or overstates the shareholders' return whenever OCI is non-zero. (b) is the numerator of ROI/RONA, an entity-level measure that excludes financing, unusual items and taxes. (d) is the after-tax operating profit used inside the Penman decomposition — a component of ROE, not ROE itself.

MCQ 2 — Correct: (d). (d) is false — DuPont is computable under any GAAP; its problems are analytical, not regulatory. (a), (b) and (c) are issues 1–3 of §7.3; the remaining two are the merely implicit cost of debt in EBT/EBIT and the absence of any place for unusual items.

MCQ 3 — Correct: (c). (c): the second adjustment nets financial assets against financial liabilities (and financial income against financial expenses), producing NFO/NE. (a) is the base formula; (b) is the first adjustment only; (d) never appears as a leverage term — operating liabilities act through the operating liability leverage inside the operating return, not through the financial multiplier.

MCQ 4 — Correct: (b). (b): with Net Financial Assets the sign in the formula flips to minus — the return on financial assets is usually lower than the operating return, so the spread is subtracted. (a) applies only with net debt. (c) is wrong: the term NFA/NE multiplies the

spread. (d) confuses this case with negative NOIC, where it is the ratio NOPAT/NOIC that loses meaning.

MCQ 5 — Correct: (a). (a): $0.70 = 1 - 30\%$ tax rate, $1.35 = \text{EBT inflated by } 35\%$ of unusual items; their product, 0.94 , looked innocuous precisely because a recurring drag and a non-recurring boost offset. (b) the rate was known (30%). (c) no such requirement exists. (d) unusual items were $+50$ before tax — positive.

True / False

T/F 1 — TRUE. Total Assets / Common Equity = (Common Equity + Total Liabilities)/Common Equity = $1 + \text{TL/CE}$. The critical part is “Total Liabilities”: the multiplier mixes operating and financial liabilities — which is precisely DuPont’s issue n. 3.

T/F 2 — TRUE. Implicit Interests = Operating Liabilities $\times$ short-term borrowing rate; ROOA = (NOPAT + Implicit Interests)/gross Operating Invested Capital. The critical part is “added back”: suppliers’ financing is not free, its cost is hidden in operating costs, and the add-back makes the gross return comparable with the borrowing rate.

T/F 3 — FALSE. The condition is not profitability but a positive spread: operating return above the cost of debt. A profitable firm (positive operating income) with a 5% return and an 8% cost of debt sees ROE fall with every unit of leverage — scenario C of Table 7.3 (-6.0%). The critical part is “always... as long as profitable”: leverage amplifies the spread, whatever its sign, and borrowing itself tends to raise the cost of debt and lower marginal operating returns.

T/F 4 — TRUE. The equivalence holds exactly under that consistency condition; the initial investment is excluded from the EVA stream because the yearly capital charge (WACC $\times$ beginning capital) already remunerates it. The critical part is “consistent assumptions”: break the link between capital evolution and cash flows and the two values diverge.

Exercise 7.4 (Real-world interpretation)

(a) With negative NOIC the ratio NOPAT/NOIC is arithmetically computable but economically meaningless: a positive profit divided by negative capital produces a negative “return” that signals nothing about performance — this is the second extreme situation of §7.8, the Microsoft 2005 configuration (NOA of -640 against NFA of $48,755$). Operating liabilities more than finance the entire operating investment, so there is no net operating capital on which to “earn a return”.

(b) Present the operating performance on the gross capital: ROOA = (NOPAT + implicit interests)/gross Operating Invested Capital, where the denominator is positive and economically interpretable, and disclose the operating liability leverage separately. Then apply the Net Financial Assets variant of the ROE formula (§7.8.1), in which the return on the

35,000 of financial assets is subtracted from the operating component with the NFA/NE multiplier of roughly 1.04.

(c) Three questions. First, capital allocation: equity holders finance a giant portfolio of financial assets whose return is almost certainly below the operating return — every year of retention dilutes ROE (the §7.8.1 logic). Second, distribution policy: the Microsoft precedent shows the natural resolution — massive returns of capital, which shrink equity and mechanically raise ROE without touching operations. Third, sustainability of the negative NOIC: the operating model relies on customers and suppliers financing the business (deferred revenues, payables); the analyst must test how that spontaneous financing behaves under stress.

Chapter 8 — The Statement of Cash Flows: Purpose, Structure, and Preparation

8.8.1 Numerical exercises

Solution to Exercise 8.1 — Monetary income and cash flow from operating activities

Step 1 — Monetary income. Starting from net income, we add back non-monetary costs (depreciation and write-downs), subtract non-monetary revenues (the equity-method revaluation), and reverse the monetary items that belong to investing activities (the gain is subtracted, the loss is added back):

Item	Amount
Net income	4,000
+ Depreciation of PP&E (non-monetary cost)	6,000
+ Write-downs of PP&E (non-monetary cost)	800
– Revaluation of financial assets (non-monetary revenue)	(300)
– Gain on disposal of PP&E (reclassified to investing)	(1,200)
+ Loss on disposal of financial assets (reclassified to investing)	400
Monetary income	9,700

Step 2 — Change in NOWC. An increase in operating assets absorbs cash; an increase in operating liabilities releases cash:

Item	Effect on cash
– Increase in net accounts receivable	(2,500)
+ Decrease in inventory	1,000
+ Increase in accounts payable to suppliers	1,800
+ Increase in product warranty reserve	100
– Decrease in tax liabilities	(300)
Change in NOWC (and tax liabilities)	100

Step 3 — Cash flow from operating activities. $CFO = 9,700 + 100 = 9,800$.

Economic meaning. VELA converts a profit of 4,000 into an operating cash flow of 9,800. The difference is driven mainly by 6,800 of non-monetary charges: the business consumed fixed assets recognized as expenses without any cash outflow in the period. The working-capital effect is almost neutral (+100): the cash absorbed by growing receivables was offset by lower inventory and longer supplier payment terms.

Common mistakes. (i) Adding, instead of subtracting, the gain on disposal — the gain is monetary, but it belongs to investing activities; (ii) treating the equity-method revaluation as

a cash inflow — it is a pure valuation entry; (iii) reversing the signs of working-capital changes (an increase in receivables reduces cash, it does not increase it).

Solution to Exercise 8.2 — Reconstructing the cash flows from investing activities

Step 1 — Payments for investments. The cash actually paid equals the opening payables (settled during the year), plus the purchases of the year, minus the closing payables (not yet paid): $1,500 + 12,000 - 4,000 = 9,500$ (outflow of $-9,500$).

Step 2 — Proceeds from the divestment. The sale price is the net book value of the asset sold plus the realized gain: $(8,000 - 5,000) + 700 = 3,700$. Since 500 has not yet been collected, the cash proceeds are $3,700 - 500 = 3,200$.

Step 3 — Net cash flow. Net investing cash flow on PP&E = $-9,500 + 3,200 = -6,300$.

Economic meaning. The accrual figures (purchases of 12,000, historical cost sold of 8,000) tell us about investment decisions; the cash figures ($-9,500$ and $+3,200$) tell us when those decisions hit liquidity. The growing balance of equipment-supplier payables ($+2,500$) financed part of the CAPEX, exactly as in the CANNON case.

Common mistakes. (i) Using the historical cost (8,000) instead of the net book value (3,000) to determine the sale price; (ii) ignoring the change in payables to equipment suppliers, thereby confusing the purchase with the payment; (iii) forgetting to deduct the uncollected receivable from the proceeds.

Solution to Exercise 8.3 — The composition of cash flow from financing activities

Step 1 — Proceeds from the capital increase. Of the 10,000 subscribed, 2,000 is a bonus issue (a pure reclassification from reserves to share capital, with no cash effect) and 3,000 has not yet been paid in by shareholders. Cash proceeds = $10,000 - 2,000 - 3,000 = 5,000$.

Item	Amount
Repayment of the current portion of M/L financial debt	(2,000)
New medium- and long-term financial loans	9,000
Change in short-term financial loans	(4,000)
Proceeds from capital increase transactions	5,000
Dividends paid	(1,500)
Cash flow from financing activities	6,500

Economic meaning. The financing section is positive ($+6,500$): the company is a net raiser of capital, lengthening its debt maturity (new M/L loans replace short-term debt) and strengthening equity. An analyst would ask what these resources are funding — growth investments would be reassuring; a structurally negative operating cash flow would not.

Common mistakes. (i) Including the stock dividend among cash inflows — it is a significant noncash financing transaction, to be disclosed separately; (ii) treating the full subscribed amount (10,000) as a cash inflow despite the unpaid portion; (iii) classifying the repayment of equipment-supplier payables here — that belongs to investing activities.

8.8.2 Multiple-choice questions

Solution to Question 1 (correct answer: B)

B is correct. The gain is a monetary item, but it arises from a divestment, not from recurring purchase–conversion–sale transactions. Removing it from operating income and adding it to the net book value in the investing section reconstructs the full disposal proceeds in one place — and prevents companies from artificially inflating operating cash flow with disposal gains.

A is wrong because the gain does have a cash counterpart (it is part of the sale price collected); the adjustment is a reclassification, not the elimination of a non-monetary item.

C is wrong because the related cash flow is not excluded from the statement: it is presented within investing activities.

D is wrong because disposal gains are regularly recognized in the income statement; the issue is only where the corresponding cash flow is classified.

Solution to Question 2 (correct answer: A)

A is correct. A three-month Treasury bill is a short-term, highly liquid investment, readily convertible into a known amount of cash and subject to insignificant risk of changes in value — the defining features of a cash equivalent.

B is wrong because equity securities are exposed to significant price risk: as noted in the CANNON case, the concept of cash and cash equivalents does not include marketable equity securities, even when they are short-term investments.

C is wrong because an original maturity of twelve months exceeds the conventional three-month threshold for cash equivalents.

D is wrong because a trade receivable is not an investment of surplus liquidity; it is an operating asset whose change affects the operating section of the statement.

Solution to Question 3 (correct answer: B)

B is correct. Payables outstanding at year-end represent costs recognized but not yet paid: relative to the expenses in the income statement, the company has retained cash. This is why the increase is added in the NOWC adjustment (in CANNON, +12,050).

A is wrong because it reverses the sign: it is a decrease in payables (payments exceeding new purchases on credit) that absorbs cash.

C is wrong because trade payables arise from operating costs of ordinary business activity, not from loans: they belong to operating, not financing, activities.

D is wrong because every change in an operating asset or liability linked to monetary revenues and costs affects the conversion of income into cash.

Solution to Question 4 (correct answer: B)

B is correct. Earnings are influenced by estimates and judgments, whereas cash flows are generally less susceptible to manipulation: a persistent gap between positive net income and negative operating cash flow is a classic signal that profits are not being converted into liquidity, and it warrants an investigation of receivables, inventories, and provisions.

A is wrong because the typical mature-company profile is the opposite: positive operating cash flows, negative investing flows, and financing outflows.

C is wrong because the entire premise of this chapter is that profitability and liquidity are distinct concepts: a profitable company can fail for lack of cash.

D is wrong because the divergence is a red flag, not proof of fraud: legitimate explanations (e.g., rapid growth absorbing working capital) exist and must be tested before concluding.

Solution to Question 5 (correct answer: C)

C is correct. Dividends are a distribution of resources to the providers of equity capital: they are a financing outflow (in CANNON, -1,000).

A is wrong because, in the classification adopted in this chapter, interest and dividends collected are operating inflows (Table 8.1); reclassifying financial income to investing is a voluntary, discretionary choice of some analysts.

B is wrong because payables to equipment suppliers arise from investments: their settlement is part of the payments for investments in PP&E, within investing activities.

D is wrong because purchases of securities are investments of liquidity: they belong to investing activities.

8.8.3 True or false?

Statement 1 — FALSE. The two methods differ only in the presentation of the operating section: the direct method shows gross receipts and payments, the indirect method reconciles net income to cash. Although the presentation differs, both methods produce the same total operating cash flow. The critical word is “different totals”: it is the format, not the amount, that changes.

Statement 2 — FALSE. Keeping goods in inventory means investing cash in those stocks: an increase in inventory absorbs financial resources and is subtracted in the NOWC adjustment. The critical part of the statement is the direction of the change: it is a decrease in inventory that frees up cash (in CANNON, the reduction of 3,000 was a source of liquidity).

Statement 3 — FALSE. A bonus issue converts reserves into share capital: no cash changes hands. Only paid-in capital increases that have actually been paid generate a financing inflow — which is why, in CANNON, the subscribed increase of 15,000 produced cash proceeds of only 7,000 (12,000 paid-in, less 5,000 of subscriptions still receivable, with 3,000 as stock dividend).

Statement 4 — TRUE. This is precisely the limited exception to the gross-reporting principle: when there is a trading perspective with potentially hundreds of buying and selling transactions, presenting the net change (in CANNON, a net investment of -11,250) is permitted and more informative than an unreadable list of gross flows.

8.8.4 Real-world interpretation exercise

Solution to Exercise 8.4 — Reading CANNON like an analyst

(a) Inherent financial equilibrium. The recurring source — cash flow from core operating activities of +27,250 — comfortably covers the recurring needs: payments for financial expenses (2,300), income taxes (3,000), the repayment of maturing loan installments (3,500), dividends (1,000), and the net investments in PP&E (6,500), which in this case can reasonably be read as maintenance CAPEX. The Operating Free Cash Flow of +20,750 confirms that the equilibrium does not depend on non-recurring sources: the disposals of fixed assets (9,500 + 8,000) and the capital increase were not necessary to cover recurring needs. The verdict is positive.

(b) The financing strategy. The company is rebalancing its financial structure: expensive and volatile short-term factoring debt is cut by 13,000 and replaced with 12,000 of new medium- and long-term loans, while equity is strengthened with 7,000 of fresh capital. Maturities are lengthened and refinancing risk falls. The excess liquidity generated was partly invested in marketable securities (11,250) and partly accumulated as cash (the balance rises from 250 to 9,850): the company moved from a fragile liquidity position to a comfortable buffer. A credit analyst would read this year very favorably.

(c) What to monitor. Three items deserve attention. First, net accounts receivable grew by 10,800 against sales of 90,000: if commercial terms are being stretched to support sales, future collections — and the allowance for doubtful accounts, already increased by 1,200 — are at risk. Second, the increase in trade payables (+12,050) released cash this year, but supplier financing cannot expand indefinitely: the positive NOWC contribution (+3,170) may reverse. Third, the litigation settlements (-2,700) and the use of provisions are non-recurring by nature:

the analyst must check whether new disputes are accruing. In short: this year's excellent cash conversion partly rests on working-capital movements that must be verified for sustainability over a multi-year horizon.

Chapter 9 — The Statement of Cash Flows under IAS 7

A. Numerical solutions

Solution A.1.

Operating profit	2,000
Add: depreciation (non-cash)	300
Less: increase in trade receivables	(250)
Add: decrease in inventories	120
Add: increase in trade payables	80
Cash from operations before taxes	2,250
Less: income taxes paid	(400)
Net cash from operating activities	1,850

Reasoning: start from operating profit, add back the non-cash depreciation, then adjust for working capital. A receivables increase is a use of cash (sales not yet collected); an inventory decrease releases cash; a payables increase is a source of cash (purchases not yet paid). Taxes paid are subtracted in full.

Common mistake: reversing the working-capital signs — adding the receivables increase instead of subtracting it. Always ask “did this change put cash in or take cash out?” An asset going up consumes cash; a liability going up provides it.

Solution A.2.

Net change from the three activities = $1,200 - 900 + 50 = +350$. Adding this to the opening 500 gives 850 before the FX effect. The unrealised translation loss of 30 is not a cash flow; it is shown as a separate reconciling line below the three subtotals. Closing cash = $500 + 350 - 30 = 820$.

Cash at beginning of year	500
Net increase from operating, investing, financing	350
Effect of exchange-rate changes on cash (non-cash)	(30)
Cash at end of year	820

Common mistake: treating the FX loss as an investing or operating outflow, which would both misstate the activity subtotals and imply a cash movement that never occurred.

Solution A.3.

(a) The investing section shows the cash price net of cash acquired: $1,000 - 120 = 880$ (an outflow of 880).

(b) No. The target's PP&E (700), inventories (150) and payables (90) are consolidated into the closing balance sheet but generate no cash flow; they never appear in the statement.

(c) Because the 700 of PP&E that the firm effectively acquired is hidden inside the single 880 acquisition line. An analyst computing capital expenditure from the "purchase of PP&E" line alone would miss it entirely and overstate free cash flow. To measure true CAPEX, the asset content of the acquisition must be added back.

Common mistake: forgetting to net off the acquired cash (reporting 1,000 instead of 880), or assuming the target's assets flow through investing line by line.

B. Multiple-choice solutions

B.1 Correct answer: (b). The amended IAS 7, aligned with IFRS 18, requires the operating profit or loss subtotal as the starting point.

Why the others are wrong: (a) net income is the old pre-amendment starting point; (c) gross profit is too high up the income statement and ignores operating expenses; (d) profit before tax includes financing items that are not operating.

B.2 Correct answer: (c). IAS 7 expressly allows overdrafts that are integral to cash management to be a (negative) component of cash and cash equivalents.

Why the others are wrong: (a) is the general rule for bank borrowings but not for integral overdrafts; (b) investing is irrelevant to a financing-type instrument; (d) the overdraft is reflected, just inside the cash definition.

B.3 Correct answer: (c). Depreciation reduced operating profit but consumed no cash, so it is added back to undo a non-cash deduction.

Why the others are wrong: (a) it is not itself a source of cash; the add-back merely reverses a non-cash charge; (b) it reduces, not increases, taxable income; (d) it is not a financing flow.

B.4 Correct answer: (c). For an ordinary entity the default classifies interest paid as financing.

Why the others are wrong: (a) operating and (b) investing are not the default for an ordinary entity's interest paid; (d) interest paid is a genuine cash flow, not a non-cash item.

B.5 Correct answer: (c). Only the cash consideration net of cash acquired appears (in investing); the non-cash assets and liabilities produce no cash flow.

Why the others are wrong: (a) they do not appear gross; (b) they are not operating; (d) they are not financing.

C. True / False solutions

C.1 False. The critical word is “always.” Short-term marketable securities are normally classified outside cash and cash equivalents, as financial assets at fair value through profit or loss. Cash equivalents require insignificant risk of value change and very short maturity.

C.2 False. An unrealised FX gain raises net income but is exactly offset by the increase in the foreign-currency receivable, so operating cash flow is unchanged. The cash has not been collected.

C.3 True. Taxes paid are presented in operations unless a flow can be specifically identified with an investing or financing transaction. The default is the operating section, and the full amount paid is reported there.

C.4 False. This is precisely the manipulation IAS 7 blocks. Only the cash price net of cash acquired enters investing; the target’s payables are non-cash and never touch the operating section. The crude trick does not work under the standard.

D. Real-world interpretation — model answer

D.1 Operating profit is flat-to-rising (500 → 560), but operating cash flow collapses (480 → 300 → 150). The widening wedge between profit and cash is the central signal: reported earnings are increasingly not converting into cash. The stated cause — sharply rising receivables and inventories — confirms that working capital is absorbing the profit. From an equity perspective this is an earnings-quality red flag: profit growth that is not cash-backed is fragile and may reverse, and one should ask whether revenue is being recognised aggressively or sales pushed to weaker customers. From a credit perspective the concern is liquidity and debt service: operating cash flow of 150 is a thin cushion, and the two acquisitions — each buried in a “net of cash acquired” line — may hide both real capital deployment and new debt, so true free cash flow is likely worse than it appears. Questions to raise: what drove the receivables and inventory build; how were the acquisitions funded; and what is free cash flow once acquisition-embedded CAPEX is added back?

Common mistake: accepting the CFO’s framing — “record operating profit” — at face value and overlooking that the cash trend is the opposite of the profit trend.

Chapter 10 — Reformulating and Analysing the Cash Flow Statement

Numerical Exercises

Exercise 1 — From CFOA to OFCF

Sigma SpA — reformulated (change in Cash), € thousands	Amount
EBITDA	45,000
+/- Change in NOWC	(6,000)
= Operating Cash Flow (OCF)	39,000
– CAPEX	(20,000)
= Operating Free Cash Flow (OFCF)	19,000
+ Interest & dividends received	1,200
– Interest paid	(4,800)
– Income taxes paid	(7,000)
– Net purchase of long-term financial assets	(3,000)
+ Net new borrowings	9,000
+ Equity issued – Dividends paid	(3,000)
= Net Change in Cash & Cash Equivalents	11,400

(a) $OCF = 45,000 - 6,000 = €39,000k$; $OFCF = 39,000 - 20,000 = €19,000k$; net change in cash = $19,000 + 1,200 - 4,800 - 7,000 - 3,000 + 9,000 + 2,000 - 5,000 = €11,400k$.

(b) CFOA starts from all income-statement items, so it embeds interest received (+1,200), interest paid (–4,800) and taxes (–7,000): $CFOA = 39,000 + 1,200 - 4,800 - 7,000 = €28,400k$. The gap versus OCF (€39,000k) is €10,600k — precisely the financial and tax flows that OCF excludes. Treating them as equal would misstate the operating engine by €10,600k.

(c) $DSCR = OCF / (\text{interest} + \text{principal}) = 39,000 / (4,800 + 8,000) = 39,000 / 12,800 = 3.05\times$. The recurring operating cash flow covers next year's debt service roughly three times over — comfortable — provided the €39,000k OCF is itself recurring (no one-off working-capital release or unusual collection inside it).

Exercise 2 — Cash vs. Non-Cash Change in Net Financial Position

(a) Other Cash Flows (B) = $800 - 3,200 - 4,000 - 18,600 + 5,000 - 6,400 = -26,400$. Cash-based reduction in net debt = $OFCF + B = 22,000 - 26,400 = -4,400$. In other words, the cash flows over the year increased net debt by €4,400: the positive OFCF of €22,000 was more than absorbed by financial investing and by financing/dividend outflows.

(b) Total change in NFP = $68,000 - 50,000 = +18,000$ (net debt up 18,000). Split it: cash-based change = +4,400 (from part a); non-cash change = $18,000 - 4,400 = +13,600$. The non-cash component reconciles exactly to the two disclosed items — the €12,000 finance lease (+12,000) and the €1,600 FVOCI revaluation (+1,600) — which together add €13,600 of net debt that never appeared in the cash flow statement. The reconciliation therefore ties out: $50,000 + 4,400 + 12,000 + 1,600 = 68,000$.

(c) For the credit committee: even on a cash basis Delta's net debt rose (€4,400), but the far larger story is the €13,600 of non-cash debt — overwhelmingly a finance lease taken instead of buying the machinery outright. A finance lease is debt in economic substance; a covenant tested on the cash bottom line alone would miss it entirely and record a much smaller deterioration. The committee should size headroom against the NFP reconciliation, not the cash statement.

Exercise 3 — Recurring Operating Cash Flow and the Quality-of-Earnings Ratio

(a) Reported Quality-of-Earnings = $21,000 / 30,000 = 0.70$. Below 1 — a first flag that profit is not fully converting into cash — but the raw figure is contaminated by an unusual inflow and an unusual provision, so it cannot be read at face value.

(b) Remove the €9,000 unusual litigation collection: it belongs among Other Cash Flows, not operating. Remove the €4,000 unusual restructuring provision increase from ΔNOWC (the “NOWC trick”): as an unusual liability it should not sit in current operating liabilities, so its €4,000 non-cash add-back must be reversed out of the recurring operating cash flow. Recurring Operating Cash Flow = $21,000 - 9,000$ (unusual inflow out) $- 4,000$ (unusual provision add-back reversed) = €8,000k. Note both adjustments cut the operating figure: one strips a genuine but non-recurring cash inflow, the other strips a non-cash accrual that had propped it up.

(c) The recurring operating cash the analyst should compare with recurring earnings is far below reported CFOA. Against a recurring net income also stripped of the litigation gain's income effect ($30,000 - 9,000 = 21,000$), the recurring ratio is $8,000 / 21,000 \approx 0.38$ — against a reported 0.70 — and the recurring conversion of profit into cash is weak. The lesson mirrors Firm B in the opening problem: the headline 0.70 already looked soft, but the recurring picture is materially worse — the firm's core operations are converting little profit into cash, and any coverage ratio built on reported CFOA would badly overstate debt-service capacity.

Multiple-Choice Questions

MCQ 1. Correct answer: B — CFOA is generated by all income-statement items (indirect method starts from net income), so it embeds financial and tax flows; OCF is generated only by operating items (starts from operating income). A confuses presentation with content; C is false (CAPEX is subtracted after OCF to get OFCF); D is the exact confusion this chapter warns against.

MCQ 2. Correct answer: B — By the rule “if something is inside the bottom line it cannot stay above the bottom line”, flows that merely move money between NFP components — short-term financial investments and financing with third parties — vanish. Interest, taxes, long-term financial investing and financing with shareholders remain (they change the NFP amount), so A, C and D stay in the statement.

MCQ 3. Correct answer: B — The NFP can change through cash transactions, internal reshuffles, and non-cash valuations/transactions. A €14m wedge between the cash-based and total change is the third channel — typically a lease of PPE recognised as financial debt. A is wrong (the two are not meant to be equal — that is why a reconciliation exists); C and D are irrelevant to the wedge.

MCQ 4. Correct answer: B — Warranties are a recurring cost of frequent sales: they belong in recurring EBITDA, their liability is a current operating liability inside NOWC, and warranty payments flow through OCF via Δ NOWC. The litigation provision is the mirror image (out of EBITDA, out of NOWC, settlement among Other Cash Flows). A and C describe the unusual provision; D is false.

MCQ 5. Correct answer: C — A ratio well below 1 (e.g. 0.5–0.6) flags that reported income is not backed by operating cash — the classic signature of aggressive estimates or revenue-recognition games. A is the opposite case (that raises the ratio); B would push the ratio above 1 given the depreciation add-back; D is unrelated.

MCQ 6. Correct answer: B — Ketz’s point: CFOA adds back D&A and the FCF then subtracts CAPEX, so the two cancel precisely when $\text{CAPEX} = \text{D\&A}$, i.e. $\text{CAPEX}/\text{D\&A} = 1$. Above 1 the firm is growing its base; below 1 it is under-investing and flattering FCF. A, C and D are unrelated to the offset.

MCQ 7. Correct answer: C — $\Delta\text{NOWC}/\text{Revenues} = (\text{NOWC}/\text{Revenues}) \times g/(1+g)$. With $\text{NOWC}/\text{Revenues}$ fixed, accelerating growth still lifts $g/(1+g)$, so the working-capital drag grows and $\text{OCF}/\text{Revenues}$ falls. A and D affect OFCF but not this specific result; B is a financial, not operating, driver.

True / False Questions

T/F 1. FALSE. The chapter’s primary definition is $\text{OFCF} = \text{Operating Cash Flow} - \text{CAPEX}$, where OCF starts from operating income and excludes interest/taxes. $\text{CFOA} - \text{CAPEX}$ is a distinct, alternative measure (Ketz); it differs from OFCF because CFOA already contains financial and tax flows. Conflating the two is the naming error flagged above.

T/F 2. FALSE. The NFP changes through three channels; only the first (cash transactions) appears in the cash flow statement. Internal reshuffles change the content but not the amount, and non-cash valuations and transactions (finance leases of PPE, fair-value/amortised-cost

changes) change the amount without any cash flow. This is why an NFP reconciliation is always required.

T/F 3. FALSE. These items have no cash flow at all — they are pure accounting valuations and are simply added back in the indirect method. Only unusual items with a genuine settlement (litigation, restructuring) have a cash flow to reclassify; gains/losses on disposals already have their cash disclosed in the Investing section.

T/F 4. TRUE. $EQI = \text{operating cash margin} - \text{net margin}$. A significant decline means cash is lagging income, so non-cash balance-sheet accounts build up — a warning about earnings sustainability. A rise means the opposite, and may signal a balance sheet being liquidated. A stable EQI signals equilibrium.

T/F 5. FALSE. The working-capital drag is $\Delta NOWC / \text{Revenues} = (NOWC / \text{Revenues}) \times g / (1+g)$, which rises with the growth rate even when $NOWC / \text{Revenues}$ is fixed. Faster growth therefore lowers $OCF / \text{Revenues}$ despite stable margins and stable working-capital efficiency — the counter-intuitive result of Section 7.2.

Interpretation Exercise (10.D) — Reading a Firm Sliding Toward Insolvency

(a) The operating engine collapses: recurring OCF falls from €60m to €35m to –€10m, so OFCF turns from +€30m to –€22m. Meanwhile the financing line balloons from €5m to €40m to €70m — the firm is increasingly funded by new debt and equity, not by operations.

(b) The cash balance rises steadily (+€23m, +€33m, +€32m), which in isolation looks reassuring. It is an illusion: the cash is coming from financing inflows (plotted above the axis), not from operations (sinking below it). Strip out financing and the firm burns cash. This is exactly the stacked-bar silhouette of the insolvency cases of Chapter 8 — operating flow shrinking, plugged by financing — and the construction example there ends in insolvency in 20X5.

(c) The Quality-of-Earnings ratio falls from 1.10 to 0.75 and by 20X5 becomes not meaningful: operating cash turns negative while income is still reported — profit with no cash behind it at all. Combined with the negative OCF, this points to income propped up by accruals — receivables or work-in-progress that are not converting — a classic construction-sector warning.

(d) Conclusion for the committee: Kappa is not a going concern on its own cash generation; it is surviving on external financing while its operating engine burns cash and its earnings quality deteriorates. Do not extend or renew unsecured exposure. The single question to management: what is driving the collapse in recurring operating cash flow — is it receivables/WIP build-up on specific contracts, and if so, are those receivables collectible, or are they the accrual residue of revenue recognised but not billed or collected?

Chapter 11 — Creative Cash Flow Reporting: Detecting Manipulation and Measuring Sustainable Operating Cash Flow

7.A Numerical Exercises

Solution to Exercise 1 — Undoing a Section Shift (Vela SpA)

Vela SpA — from reported CFOA to reformulated operating cash flow	€ m
Reported Cash Flow from Operating Activities	140
– Receivable sale reclassified to Financing	(30)
– Capitalised recurring development spend, now operating	(18)
– Securities-trading inflow reclassified to Investing	(12)
= Reformulated (sustainable-basis) Operating Cash Flow	80

Working: the €30m factoring inflow is a financing loan, removed from operations. The €18m of recurring development cost belongs in operations as an ordinary outflow, so it is subtracted; the €6m amortisation add-back requires no adjustment, because it is a non-cash item that merely offsets the €6m amortisation expense in net income — its net effect inside CFOA is zero. The €12m of securities-trading cash is investing in economic terms. Reported CFOA of €140m collapses to €80m once the three section-shifts are undone — a €60m gap, and a very different signal about the operating engine.

Interpretation: an equity analyst valuing Vela on €140m of operating cash would overstate the recurring base by nearly half; a credit analyst sizing a facility on it would lend against cash that is partly a disguised loan and partly one-off securities and capitalisation effects.

Solution to Exercise 2 — Non-Recurring Items and Sustainable OCF (Corvo SpA)

Sustainable OCF = 410 + 22 (restructuring, a non-recurring outflow → add back) + 15 (pension top-up, non-recurring outflow → add back) – 19 (litigation receipt, non-recurring inflow → subtract) = €428m. The €28m of ordinary operating expenses is recurring and is not adjusted. The logic is the symmetry of §5.1: reverse the cash effect of anything that will not repeat. Two of the items had depressed reported OCF (so they are added back) and one had inflated it (so it is subtracted); the net effect lifts the sustainable figure to €428m, €18m above the reported number, because the non-recurring outflows outweigh the non-recurring inflow.

Solution to Exercise 3 — Growth vs. Profitability and Efficiency (Rondine SpA)

Year-1 OI-to-sales = $40\% - 25\% = 15\%$. Increase in sales = €2m.

Effect	Formula	Amount	Cash
Growth on the OI	$\text{€}2\text{m} \times 15\%$	€300,000	Source
Growth on working capital (OWC)	$\text{€}2\text{m} \times 12\%$	€240,000	Use
Change in profitability (OI)	$\text{€}22\text{m} \times 1\%$	€220,000	Use
Change in efficiency (OWC)	$\text{€}22\text{m} \times 3\%$	€660,000	Use

Reading it: growth on the cushion is the only source (+€300,000). Growth on working capital ties up €240,000; the 1-point slip in the OI ratio costs €220,000; and the 3-point rise in OWC-to-sales costs €660,000. Net, the year's changes are a use of roughly €820,000 of core operating cash despite €2m of extra sales — the growth cash-flow profile turned sharply negative because both profitability and efficiency deteriorated at once. The single question for management mirrors Chapter 10: is the working-capital build-up (the €660,000 efficiency drag) a deliberate investment in growth or an early sign of uncollectable receivables and slow-moving inventory?

7.B Multiple-Choice Questions

MCQ 1. Correct answer: D — Profitability does not guarantee positive cash flow: a growing, profitable firm whose receivables and inventory are expanding can post negative operating cash flow (the working-capital drag of §5.2). A, B and C are all accurate.

MCQ 2. Correct answer: B — EBITDA is an earnings-based measure, not a true cash-flow measure: it is built on accruals and ignores working-capital movements, so it only approximates operating cash flow. A, C and D correctly describe it.

MCQ 3. Correct answer: C — Both the trading of securities and the purchase of inventory are reported in the operating section under the standards; only non-trading investment securities are investing. This is exactly why analysts re-classify securities trading to investing when isolating the operating engine (§2.4).

MCQ 4. Correct answer: B — The inventory arrives as part of the acquisition (investing outflow), but its sale is an operating inflow — the mismatch that overstates operating cash flow and lets a serial acquirer harvest inherited working capital (§3.1).

MCQ 5. Correct answer: A — Capitalising defers the expense, so net income is higher; and the cash outflow is moved to the investing section, so operating cash flow is higher than under expensing. This is the mechanism of §2.3 (shifting operating outflows to investing).

MCQ 6. Correct answer: C — Add back the two non-recurring outflows: $379 + 23 + 18 = \$420\text{m}$. The \$25m of ordinary operating expense is recurring and is not adjusted (§5.1).

MCQ 7. Correct answer: C — Under the trade-payable classification, the bank's payment to the supplier does not touch the buyer's cash, so it is a non-cash event; only the buyer's later payment to the bank is an operating outflow. Under the borrowing classification the answer would differ (operating outflow + financing inflow). This substance test is the §4.2 point.

MCQ 8. Correct answer: C — When gross profit exceeds operating expense, growth scales up a positive change of OI — a source of cash. Positive operating working capital means growth ties up receivables and inventory faster than payables — a use of cash. This is the growth cash-flow profile of §5.2.

7.C True / False Questions

T/F 1. FALSE. The opposite is Mulford and Comiskey's foundational point: the balance in cash and the total change in cash are readily verifiable against banks and hard to misstate, whereas the split across operating, investing and financing is flexible — and therefore manipulable. Creative cash flow reporting lives in the split, not the total (§1.1).

T/F 2. TRUE. The cash comes from a bank against the receivable — a financing inflow (a loan) — not a genuine collection from a customer. Reading the decrease in A/R as an ordinary working-capital release inflates operating cash flow; the fall should be excluded from working capital and classified as financing (§2.2; the Enron self-dealing case is the extreme version).

T/F 3. FALSE. The operating section shows the annual employment cost added straight back (net-zero, being non-monetary), and financing shows only the strike proceeds. The in-the-money spread at exercise — €4,000 in the worked example ($\text{€}5 \times 800$ shares) — appears nowhere. The statement understates the true cost by that hidden value transfer to the option holders (§4.1).

T/F 4. FALSE. The growth cash flow profile measures only the effect of growth on the operating income and on operating working capital. Even with a negative profile, a firm can still produce positive core operating cash by improving profitability or efficiency (changing the underlying fundamentals). It is also true, separately, that growth alone can drive core operating cash negative (§5.2).

T/F 5. FALSE. Stretching payables releases cash into operations once, by increasing a liability, but it is not sustainable: suppliers eventually refuse the slower terms (or withdraw credit), and the release cannot repeat. It is one of the four unsustainable boosts of §3.4 and must be normalised out when measuring sustainable operating cash flow.

7.D Real-World Interpretation Exercise — Reading a Serial Acquirer's Cash Flow

(a) Four practices, three families. Collecting acquired receivables and selling acquired inventory is Case 1 (§3.1): inherited working capital harvested as a one-off operating inflow. Stretching group payables is an unsustainable boost (§3.4): a one-time working-capital release. Capitalising customer-acquisition commissions is Case 2 / §2.3: a recurring operating outflow shifted to investing. And the roll-up structure itself is the serial-acquirer pattern (§3.1): growth without the initial operating outflows organic growth would require.

(b) Each pushes sustainable OCF below the reported €300m. Removing the harvested inherited working capital strips a non-recurring inflow; normalising payables to a sustainable days-payable removes another non-recurring inflow; re-classifying capitalised commissions into operating adds a recurring outflow. The direction is unambiguous even before precise figures: the sustainable number is materially lower, and — because the inherited-working-capital and payable effects cannot repeat at the same scale without ever-larger acquisitions — it may be flat or declining once the acquisition pace slows.

(c) Conclusion for the committee: Falco's reported operating cash is substantially a by-product of its acquisition machine and of temporary working-capital releases, not of a self-sustaining organic engine; the reported €300m overstates recurring operating cash and should not be extrapolated. The single question to management: excluding the receivables and inventory inherited from acquisitions and holding days-payable constant, what was organic operating cash flow last year — and what happens to it if the acquisition pace halts?

Chapter 12 — Strategic Business Analysis, Business Models, and Key Performance Indicators

A. Numerical Exercises

Exercise 12.1 — Rebuilding Ryanair's income from its KPIs

Solution.

(a) Total Revenue per Booked Passenger = average fare + ancillary revenue per passenger = $50,66 + 23,94 = €74,60$, exactly the disclosed figure. Total operating revenues = 208,4 million passengers $\times €74,60 \approx €15,55$ billion, which reconciles (up to rounding in the disclosed per-passenger figures) with the €15,54 billion reported in Table 12.7. The reconstruction demonstrates the central message of Section 12.7: for this business model, revenue is fully explained by two KPIs — passengers and revenue per passenger — so any revenue forecast is, implicitly, a forecast of those two drivers.

(b) Operating income per passenger = $74,60 - 63,21 = €11,39$. Operating margin = $11,39 / 74,60 = 15,3\% \approx 0,15$, as disclosed. Note the analytical convenience of per-passenger economics: the entire income statement is compressed into a spread between two unit values, exactly as a retailer's economics can be compressed into margin per unit $\times$ units sold.

(c) Break-Even Load Factor = $CASM / Yield = 0,0758 / 0,0946 = 0,801 \approx 80\%$. Economic meaning: since operating costs accrue on the capacity offered (ASM) while revenues accrue on the capacity sold (RPM), the airline covers its operating costs when the fraction of seats sold reaches 80%. With an actual load factor of 93,8%, Ryanair flies almost 14 percentage points above break-even — a substantial safety margin. Observe from Table 12.6 how this margin collapsed in FY2022 (load factor 82,2% vs. break-even 88%): the airline was operating below break-even, consistent with the negative operating margin of (7)%.

(d) $ASM = \text{seats available} \times \text{average sector length} = 220,0 \text{ million} \times 790 \text{ miles} \approx 173,8 \text{ billion} \approx 174 \text{ bn}$, as disclosed. $RPM = \text{passengers booked} \times \text{average sector length} = 208,4 \text{ million} \times 790 \approx 164,6 \text{ billion} \approx 164 \text{ bn}$. As a cross-check, $RPM/ASM = 164/174 = 94,3\%$, consistent with the disclosed load factor of 93,8% (the small difference arises because the average individual aircraft capacity in Table 12.7 is rounded to 191 seats: the disclosed load factor of 93,8% implies about 222,2 million seats, i.e. roughly 193 seats per sector, and Ryanair computes the disclosed load factor on unrounded data).

Common mistakes: forgetting ancillary revenues when reconstructing revenue per passenger (they are almost one third of the total, and the most stable component across the five years); computing the break-even load factor as costs/revenues (that is a cost-income ratio, not a load factor); mixing capacity measures (ASM) with traffic measures (RPM) in the denominator of unit costs.

Exercise 12.2 — Testing a hotel business plan (the three-step test of Section 12.7)

Solution.

Step 1 — actual ratios to actual KPIs. Available room-nights = $1.000 \times 365 = 365.000$. Occupied nights = $365.000 \times 71\% = 259.150$. Revenue check: $259.150 \times \text{€}145 \approx \text{€}37,6$ million — the actual KPIs explain the actual revenue. $\text{RevPAR} = \text{ADR} \times \text{occupancy} = 145 \times 0,71 = \text{€}102,95$.

Step 2 — implied projected KPIs. Projected $\text{RevPAR} = 48,1 \text{ million} / 365.000 = \text{€}131,8$. With an ADR of $\text{€}155$, the implied occupancy is $131,8 / 155 = 85,0\%$. The plan therefore requires occupancy fourteen percentage points above the current level and nine points above the best-in-class competitor (76%).

Step 3 — judgment. The projected financial figures are internally consistent, but the operational KPI beneath them is not reasonable: no evidence supports occupancy above the best performer in the segment. The analyst should either reject the plan or ask management which combination of drivers really carries the growth (higher ADR? mix shift? new demand segments?), and then re-test. Note the diagnostic power of the method: the plan looked plausible in euros; it became implausible only when translated into KPIs.

Common mistakes: checking only the arithmetic of the plan (internal consistency) without benchmarking the implied KPIs against the industry (external reasonableness); comparing revenue growth without holding capacity fixed, thereby missing the occupancy constraint entirely.

Exercise 12.3 — Sensitivity analysis on operating income

Solution.

(a) Revenue per passenger = $60 + 20 = \text{€}80$. Operating income per passenger = $80 - 70 = \text{€}10$. Operating income = $10 \text{ million} \times \text{€}10 = \text{€}100 \text{ million}$; operating margin = $10/80 = 12,5\%$.

(b) The three what-if cases are summarized in the following Table.

What-if case	New values	Operating income	Δ vs. base	Elasticity
Base case	—	€100 m	—	—
(i) Average fare -5%	fare €57 → rev/pax €77	$10 \text{ m} \times (77-70) = \text{€}70 \text{ m}$	-30%	6,0
(ii) Fuel +10%	fuel €23,1 → cost/pax €72,1	$10 \text{ m} \times (80-72,1) = \text{€}79 \text{ m}$	-21%	2,1
(iii) Passengers +5%	10,5 m passengers	$10,5 \text{ m} \times \text{€}10 = \text{€}105 \text{ m}$	+5%	1,0

Sensitivity of operating income to single-variable changes (elasticity = $\% \Delta \text{ operating income} / \% \Delta \text{ driver}$).

(c) The fare is by far the most critical driver: a 1% change in fare moves operating income by roughly 6% — the classic leverage of a thin-margin, high-volume business, where a €3 fare cut destroys €30 million of income because it falls entirely on a €10 unit margin. Fuel follows, with an elasticity of about 2. Passenger volume appears the least critical, but only because of the simplifying assumption that all costs are variable: in reality a large share of airline costs (fleet ownership, crew rosters, airport agreements) is fixed in the short run, so the true elasticity to volume is much higher on the downside — which is precisely why the load factor and the break-even load factor of Exercise 12.1 are the industry's most obsessively monitored KPIs. This is also the structural limit of one-variable sensitivity analysis: in reality drivers move together (fuel spikes depress demand; fare cuts raise load factors), which is why Section 12.8 pairs sensitivity analysis with scenario analysis and stress testing.

B. Multiple Choice Questions

MCQ 1

Correct answer: (b).

(b) is correct: as Section 12.5.5 explains, KPIs generally measure results that have already occurred, whereas KRIs are constructed to provide early-warning signals before objectives are compromised (Beasley, Branson and Hancock, 2010).

(a) is wrong because both families of indicators are quantitative metrics; the distinction concerns timing and purpose, not measurability.

(c) is wrong because KPIs and KRIs are complementary: one measures achievement, the other monitors threats to achievement — mature systems integrate both.

(d) is wrong because neither KPIs nor KRIs are standardized by accounting standards; this is exactly why the analyst must scrutinize their definitions.

MCQ 2

Correct answer: (c).

(c) is correct: tolerance operationalizes appetite as acceptable ranges around each critical KPI, triggering intervention when breached (Section 12.5.2).

(a) describes risk capacity — what the firm can bear, not what it accepts.

(b) describes risk appetite — the strategic willingness, set by senior management under board oversight (COSO).

(d) describes residual risk, an output of the risk-response phase, not a limit set ex ante.

MCQ 3

Correct answer: (c).

(c) is correct: $BELF = CASM/Yield$ is the load factor at which revenues from seats sold equal the costs of seats offered; flying below it means an operating loss — exactly Ryanair's situation in FY2022 (load factor 82,2% vs. BELF 88%, operating margin (7)%).

(a) is wrong because a load factor is only “high” relative to the company's own break-even, not in absolute terms — the industry-relativity principle of Section 12.6.

(b) is wrong because “close” is not enough: below break-even is below break-even, and with billions of ASM two percentage points translate into a large absolute loss.

(d) is wrong because BELF and the actual load factor are already normalized per unit of capacity; fleet size affects the size of the loss, not its existence.

MCQ 4

Correct answer: (b).

(b) is correct and restates the distinction of Section 12.3.3, grounded in Osterwalder and Pigneur (2010) and Teece (2010).

(a) is wrong: two differentiators can deliver value through completely different architectures (e.g., own retail vs. wholesale distribution).

(c) is wrong: revenue streams are one of nine blocks of the Canvas; the model also covers segments, resources, activities, partnerships and costs.

(d) is wrong: digital firms made the concept popular, but every enterprise — an airline, a hotel, a bank — employs a business model, explicitly or implicitly.

MCQ 5

Correct answer: (b).

(b) is correct: scenario analysis builds coherent alternative environments (optimistic, base, pessimistic) in which several drivers move together (Section 12.8.4).

(a) is wrong because sensitivity analysis changes one variable at a time, holding the others constant.

(c) is wrong in this framing: Monte Carlo is a probabilistic technique that samples from probability distributions; scenarios do not require probabilities.

(d) is wrong: back-testing validates forecasting models against realized data; it does not construct alternative futures.

MCQ 6

Correct answer: (c).

(c) is correct and summarizes the three-step test of Section 12.7 (actual ratios → actual KPIs → projected KPIs → projected ratios), applied in Exercise 12.2.

(a) is wrong because internal arithmetic consistency is necessary but trivially achievable in any spreadsheet; it says nothing about economic plausibility.

(b) is wrong — improving margins are the claim to be tested, not the test.

(d) is wrong: authority is not evidence; the analyst tests assumptions, not signatures.

C. True/False Questions

True/False 1

FALSE. The critical part is “standardized ... and audited”: most operational KPIs are neither defined by accounting standards nor subject to the same audit depth as the financial statements. This is precisely why the analyst must check definitions, perimeters and changes over time before comparing companies (Section 12.2).

True/False 2

TRUE. This is the core principle of Section 12.7: financial ratios are consequences of operational drivers. A margin that improves without a plausible KPI story (efficiency, price, mix, utilization) is an assumption, not a forecast — as Exercise 12.2 shows.

True/False 3

FALSE. The critical distinction: capacity is what the organization is able to bear without threatening its viability; appetite is what it is willing to accept. A firm may be willing to take risks it cannot afford — an unsustainable position that the analyst, especially the credit analyst, must detect (Section 12.5.2).

True/False 4

FALSE. The critical part is “non-probabilistic”: Monte Carlo samples from explicit probability distributions and returns a probability distribution of results (e.g., 90% confidence intervals for cash flow). Producing many outcomes is not what defines the classification — the use of probabilities is (Section 12.8.3).

True/False 5

TRUE. From Table 12.6: load factor 82,2% against a break-even of 88%, and an operating margin of (7)%. The KPI system and the income statement tell the same story — the internal coherence the analyst should always verify (Section 12.9).

True/False 6

FALSE. The critical part is “has already been recorded”: the purpose of KRIs and escalation triggers is exactly the opposite — to alert management before deterioration reaches the financial statements, on a scale from minor to dramatic events (Sections 12.5.5–12.5.6).

D. Real-World Interpretation Exercise

Exercise 12.4 — Thinking like an analyst on the Ryanair disclosures

Solution (guided).

(a) Sectors flown in FY2026 = 1.151,8 thousand (Table 12.6). The fleet grew from 613 aircraft (31 March 2025) to 647 (31 March 2026, Table 12.8); a reasonable average fleet is ≈ 630 aircraft. Flights per aircraft per year $\approx 1.151.800 / 630 \approx 1.828$; per day $\approx 1.828 / 365 \approx 5,0$ sectors per aircraft per day.

(b) The average sector length is 790 miles (≈ 1.270 km), i.e. roughly 1,8–2,0 block hours per sector for a 737-class aircraft. Five sectors per day would then imply ≈ 9 –10 flight hours per aircraft per day — remarkably consistent with the disclosed Average Daily Flight Hour Utilization of 9,48 hours. Two independently disclosed KPIs corroborate each other: this is the kind of internal-coherence check an analyst should always run on a disclosed KPI set.

(c) The 5,0 sectors/day figure is an average over the whole fleet and the whole year. In peak summer scheduling a short-haul low-cost aircraft can fly six or more sectors per day; the gap between peak-capable and average utilization suggests that a meaningful share of fleet-days — plausibly of the order of 15–20% — is absorbed by scheduled maintenance, winter seasonality and tactical capacity cuts (for example when fuel prices are very high). This is an estimate, not a disclosed figure: to test it, the analyst should download the monthly traffic statistics (passengers and load factor) from the company's investor site, reconstruct the monthly seasonality of capacity, and check whether winter sector counts fall by an amount consistent with the assumed grounded share. The exercise illustrates a general lesson: disclosed KPIs rarely answer the analyst's question directly, but they almost always allow the question to be bounded, cross-checked, and turned into a verifiable hypothesis.

Common mistakes: dividing sectors by the end-of-year fleet (647) instead of the average fleet, which understates utilization in a growing airline; treating the estimate in (c) as a fact rather than as a hypothesis to be verified; forgetting that utilization statistics may be computed on “in-service” days, so definitions must be read in the annual report before comparing across airlines.

Chapter 13 — Non-GAAP Measures and Street Earnings: Adjusted Performance, Analyst Definitions, and the Quality of Exclusions

9.1 Numerical Exercises

Exercise 1 — Reconstructing the gap

Solution. (a) Total add-backs = $8,648 - 3,687 = €4,961\text{m}$; these are the items management treats as unusual. (b) Gap = $8,648 - 5,520 = €3,128\text{m}$, equal to $3,128 / 5,520 \approx 56.7\%$ of net profit. (c) The gap is positive (income-increasing) and large: at the operating level the adjusted measure exceeds the statutory bottom line by an amount equal to more than half of net profit ($\approx 57\%$). A gap of this magnitude does not by itself prove opportunism, but it obliges the analyst to test whether the $€4,961\text{m}$ of add-backs are genuinely transitory. Common mistake: computing the gap against operating income rather than net profit, or forgetting that the tax and financial-result removals are part of the bridge and not 'adjustments' in the non-GAAP sense.

Exercise 2 — Adjusted-for-recurring

Solution. On the persistence test, restructuring recurring for four years fails the frequency criterion and should be re-included; the discrete plant impairment is a defensible one-off and may remain excluded; the residual 'other' is treated conservatively as recurring absent evidence to the contrary. Analyst-adjusted operating income = $8,648 - 2,600 - 861 = €5,187\text{m}$. The correction of $€3,461\text{m}$ brings the measure far closer to the IFRS net profit of $€5,520\text{m}$, reversing most of the firm's uplift. Common mistake: re-including the impairment mechanically because it is large; size is not the criterion — persistence and discreteness are.

Exercise 3 — The multiple

Solution. (a) Statutory P/E = $8.50 / 1.70 = 5.0\times$; adjusted P/E = $8.50 / 2.80 \approx 3.04\times$. (b) You must ask which earnings definition the colleague used: the $3\times$ figure is the adjusted P/E, and the same firm at the same price trades at $5\times$ on statutory EPS — a difference of roughly 40%. Accepting '3x' without the denominator risks concluding the stock is cheap on a number that embeds the firm's own income-increasing adjustments. Common mistake: quoting a multiple without stating the earnings definition behind it.

9.2 Multiple-Choice Questions

MCQ 1

Correct answer: B. The discriminating test between the information and opportunism views is the persistence of the excluded items: transitory items predict nothing, whereas recurring items relabelled as unusual predict lower future performance. A is wrong because a large but genuinely discrete impairment can be a legitimate exclusion — size is not the criterion. C is irrelevant to the

quality of the exclusion. D confuses the label with the substance; the label is precisely what the analyst must look behind.

MCQ 2

Correct answer: C. Street earnings are the data-vendor's compiled consensus 'actual', constructed on the basis used by the majority of contributing analysts and used in beat-or-miss headlines. A is the manager non-GAAP measure; B is the statutory figure; D is a forecast, not a realised 'actual'. The three are distinct and are conflated at the analyst's peril.

MCQ 3

Correct answer: B. IFRS 18 brings the measure inside the audited financial statements: a single note must explain why the measure is useful, how it is calculated, its reconciliation to the comparable statutory subtotal, and the income-tax and non-controlling-interest effect of each reconciling item. A reverses the actual shift (from outside to inside). C confuses the firm measure with the vendor consensus. D is false — income-increasing measures are not prohibited; they are disciplined through disclosure and reconciliation.

MCQ 4

Correct answer: B. Excluding losses but keeping gains produces an income-increasing measure rather than a neutral one; symmetry of adjustments is the relevant test. A is a procedural detail unrelated to symmetry. C is false — non-GAAP disclosure is permitted, not prohibited. D concerns OCI and net income (Chapter 2), not the symmetry of non-GAAP adjustments.

MCQ 5

Correct answer: B. Divergence arises because analysts exercise their own discretion — adding back items the firm keeps, or keeping items the firm excludes — so the consensus 'actual' departs from the firm's own measure. A is false: both start from the same statutory framework. C is not the structural driver. D inverts reality — the manager measure, not the Street number, is the one heading toward audit under IFRS 18; the Street number is a vendor construct.

9.3 True / False Questions

T/F 1

FALSE. Prevalence establishes that a demand for a 'core' performance measure exists, but says nothing about the quality of any particular firm's exclusions. Widespread use is consistent with both the information and the opportunism views; the discriminating evidence is the persistence of the excluded items, not the popularity of the practice.

T/F 2

FALSE. Analyst-defined adjusted earnings are independent of the firm's definition. An analyst may re-include items the firm excludes (e.g. recurring restructuring) or strip items the firm retains (e.g. amortisation of acquired intangibles). This discretion is precisely why the Street number diverges from the manager measure.

T/F 3

FALSE. The evidence runs the other way: higher exclusion levels predict lower future operating cash flows, indicating that the excluded items are not, on average, transitory. This is the empirical tension the analyst must hold against the value-relevance evidence, and it is why the persistence test is decisive.

T/F 4

TRUE. IFRS 18 moves the firm's own measure inside the audited statements, raising its discipline, but it does not constrain the adjustments individual analysts choose to make. The Street-versus-firm gap therefore narrows in quality but does not disappear.

T/F 5

FALSE. Restructuring is one-off only if it is genuinely discrete and infrequent. A firm reporting restructuring in every year — as in a sector undergoing a prolonged transition — fails the frequency criterion; such charges are closer to a recurring cost of doing business, and excluding them inflates the apparent sustainable earnings.

9.4 Real-World Interpretation Exercise

Model answer. The first paragraph should reject the treasurer's premise for cash-relevant items. Restructuring is largely a cash cost, and its recurrence for four years — in a sector undergoing a structural transition — makes it a predictable, ongoing use of cash rather than a discrete event. Adding it back inflates apparent operating profitability without altering the cash that must leave the business, so it cannot be excluded when sizing debt-service capacity. The €861m of undifferentiated 'other' expense is treated conservatively as recurring absent evidence of discreteness. The discrete plant impairment is the one defensible exclusion, but it is a non-cash item and therefore does not affect near-term coverage either way.

The second paragraph should reconstruct a cash-relevant operating figure — re-including recurring restructuring and 'other', i.e. $€8,648\text{m} - €2,600\text{m} - €861\text{m} = €5,187\text{m}$ — and note that this sits close to the IFRS net profit of €5,520m, far below the headline adjusted figure. The credit assessment should be anchored on this reconstructed number and, ultimately, on cash flow, where recurring cash costs cannot be defined away. Headline conclusion: for credit purposes the Adjusted Operating Income overstates sustainable, cash-generating capacity by roughly €3.5bn;

debt-service capacity should be assessed nearer the statutory figure. Common mistake: accepting the firm's exclusions wholesale, or treating the non-cash impairment as if it improved coverage.

Chapter 14 — Accounting for Intangible Assets: Recognition, Measurement, and Analytical Consequences

Exercise 14.1 — Recognition and initial measurement

Step 1 — Item-by-item classification.

No.	Item	Amount	Treatment	Reason
1	Purchase of patent	800	Capitalize	Acquired in a separate transaction; identifiable through legal rights; reliable cost
2	Registration legal fees	50	Capitalize	Directly attributable cost of bringing the asset to its intended use
3	Internal research	400	Expense	All research costs must be expensed as incurred
4	Qualifying development	600	Capitalize	All six development criteria satisfied; a chosen alternative exists
5	Advertising campaign	900	Expense	Internally generated brand; benefits not probable and cost not reliably measurable
6	Staff training	200	Expense	Workforce is not separable and not controlled; fails identifiability
7	Purchase of customer list	300	Capitalize	Acquired separately; separable — capable of being sold or transferred

Step 2 — Total intangible assets recognized.

$$800 + 50 + 600 + 300 = \mathbf{1,750}$$

Step 3 — Total expensed.

$$400 + 900 + 200 = \mathbf{1,500}$$

Economic meaning. Items 5 and 6 are almost certainly value-creating investments — brand awareness and workforce capability are exactly the intangibles that drive the 90% of market value identified in the Ocean Tomo study. They are expensed because neither can be measured reliably or, in the case of training, controlled. Note the sharpest contrast in the table: item 7 (a *purchased* customer list) is capitalized, while item 5 (an *internally built* customer base) is expensed. The economic resource is comparable; the accounting outcome is opposite.

Common mistakes. (a) Capitalizing advertising on the argument that it “creates future benefit” — the benefit may exist, but the recognition criteria are not met. (b) Expensing item 2 — directly attributable registration costs form part of the cost of an acquired asset. (c) Capitalizing item 3 by reclassifying it as development — the expenditure is aimed at *obtaining new knowledge* and no alternative has been chosen, so it is research by definition.

Exercise 14.2 — Useful life, amortization, and impairment

Step 1 — Useful life and annual amortization.

Asset	Useful life	Basis	Annual amortization
License	9 years	Intended use, below the 15-year legal ceiling	$900 \div 9 = 100$
Trademark	Indefinite	Unlimited legal life + demonstrated expectation of indefinite benefit	Nil — impairment test only
Software	4 years	Expected useful life	$500 \div 4 = 125$

Step 2 — Carrying amounts at the end of Year 2.

Asset	Cost	Accum. amortization	Impairment	Carrying amount
License	900	(200)	—	700
Trademark	600	—	(150)	450
Software	500	(250)	—	250
Total	2,000	(450)	(150)	1,400

Step 3 — Income statement effect in Year 2.

Item	Amount
Amortization — license	(100)
Amortization — software	(125)
Impairment loss — trademark (600 – 450)	(150)
Total charge to income statement	(375)

Economic meaning. The three assets illustrate all three rows of the logic in Table 14.1. The license and software carry *predictable* amortization: once the useful life is set, the charge is mechanical. The trademark carries no amortization at all — until, without warning, it produces a charge of 150, larger than either annual amortization figure. This is precisely the trade-off identified in section 15.4.4: indefinite-lived intangibles replace a small predictable charge with a large unpredictable one.

Common mistakes. (a) Amortizing the license over 15 years — the legal life is a ceiling, not the useful life. (b) Amortizing the trademark over some arbitrary period such as 20 years — an indefinite useful life means *no amortization*, only annual impairment testing. (c) Recording the impairment in OCI — impairment losses under the cost model go to the income statement; only *revaluation* gains go to OCI. (d) Forgetting that if the trademark's recoverable amount later recovered, a reversal would be permitted under IAS 36 — but it would not be if the asset were goodwill.

Exercise 14.3 — Comparing two firms under different R&D policies

Step 1 — As reported.

	ALPHA (US GAAP)	BETA (IFRS)
Revenue	10,000	10,000
Other operating costs	(6,000)	(6,000)
R&D expensed	(2,000)	(1,200)
Amortization of capitalized development	—	(400)
EBIT	2,000	2,400
Tax (25%)	(500)	(600)
Net income	1,500	1,800
Opening equity	5,000	5,000
Closing equity	6,500	6,800
ROE (NI / closing equity)	23.1%	26.5%

BETA appears more profitable on every measure. It is not.

Step 2 — Adjust BETA to a US GAAP basis.

Reverse the capitalization: add back the current-year capitalized amount as an expense (800) and remove the amortization of prior capitalization (400).

	BETA as reported	Adjustment	BETA adjusted
EBIT	2,400	−800 + 400	2,000
Tax (25%)	(600)	+100	(500)
Net income	1,800	−300	1,500
Closing equity	6,800	−1,200	5,600
ROE (NI / closing equity)	26.5%		26.8%

Adjusted equity must also remove the net capitalized development asset carried on the balance sheet, net of deferred tax. The net capitalized balance is $1,200 + 800 - 400 = 1,600$; net of 25% deferred tax, the equity adjustment is $0.75 \times 1,600 = 1,200$. Adjusted closing equity is therefore $6,800 - 1,200 = 5,600$, and adjusted ROE = $1,500 / 5,600 = 26.8\%$. Once adjusted, **BETA's EBIT of 2,000 and net income of 1,500 are identical to ALPHA's.**

Note that adjusted closing equity (5,600) does not equal ALPHA's 6,500, and adjusted ROE (26.8%) does not equal ALPHA's 23.1%. The reason lies in the exercise's stipulation that opening equity is 5,000 for both firms: BETA also carries an opening capitalized development balance of 1,200, and two firms with identical histories could not show both. The income statement measures reconcile exactly; the residual equity difference of 900 ($0.75 \times 1,200$) is the after-tax capitalized balance embedded in BETA's opening position — a reminder that the balance sheet carries the cumulative effect of past accounting policy even after the current year has been adjusted.

Step 3 — Comment on comparability.

The two firms are economically identical: both spent 2,000 on R&D and generated 10,000 of revenue with 6,000 of other costs. The entire reported difference — 400 of EBIT, 300 of net income, and 3.4 percentage points of ROE — is produced by the divergence between IFRS and US GAAP described in section 15.6.2.

Note also that the direction of the effect depends on the *trajectory* of spending. Here BETA capitalizes 800 and amortizes 400, so capitalization is still adding to income. Had BETA's R&D been declining, amortization of past capitalization could have exceeded current capitalization, and BETA would have looked *less* profitable than ALPHA. This is the practical content of the qualifier “with increasing R&D expenditures”.

Common mistakes. (a) Adjusting only the current-year capitalization and forgetting to reverse the amortization of prior years' capitalized costs — this overstates the adjustment. (b) Adjusting the income statement but leaving the balance sheet unchanged, producing an internally inconsistent ROE. (c) Concluding that BETA is manipulating earnings — both firms are correctly applying their respective frameworks; the non-comparability is structural, not a reporting failure.

Multiple-choice questions

Solution — Question 1 (correct answer: B)

B is correct. This restates the two alternative routes to identifiability. Either condition is sufficient; both need not hold. Separability is the operative test for assets not protected by legal rights.

A is wrong because a determinable market price relates to *measurement* under the revaluation model, not to identifiability. Most identifiable intangibles — patents, customer lists — have no active market.

C is wrong because the ability to estimate future cash flows relates to the *recognition* condition that future economic benefits be probable, which is a separate test applied after the definition is satisfied. Confusing the definition with the recognition criteria is the most common error on this topic.

D is wrong because the duration or method of generation is irrelevant to identifiability. Internally generated intangibles frequently *are* identifiable — they simply fail the reliable-measurement condition for recognition.

Solution — Question 2 (correct answer: A)

A is correct. This is exactly the set of effects listed in section 15.6.5: income is higher, equity is inflated through retained earnings, intangible assets are inflated, and the expenditure is reclassified from an operating to an investing outflow — which raises CFO while leaving total cash flow untouched. This combination is what makes capitalization hard to detect with a simple earnings-versus-cash-flow screen.

B is wrong on two counts. CFO is *higher*, not lower, because the outflow has been moved out of the operating section. And total cash flow cannot change: the firm paid out the same cash either way.

C is wrong because capitalization is not a pure balance sheet reclassification. The expenditure would otherwise have passed through the income statement, so removing it raises reported income.

D is wrong because equity is higher, not lower. The amortization charge is smaller than the expenditure that would otherwise have been recognized in full — in the DELTA case, 600 against 3,000.

Solution — Question 3 (correct answer: B)

B is correct. Revaluation gains are credited to a revaluation reserve and recognized as other comprehensive income, while revaluation *losses* are recognized as an expense in the income statement. The asymmetry is deliberate and conservative.

A is wrong and represents the specific error the standard is designed to prevent: routing unrealized upward revaluations through profit or loss would allow firms to report income from holding assets they have not sold.

C is wrong — that describes the pre-2023 US GAAP treatment of cryptocurrencies, where gains were recognized only upon sale, not the IFRS revaluation model.

D is wrong because no such rule exists. There is a genuine complication where an upward revaluation follows a previous downward one, but it concerns the *channel* of recognition, not accumulated amortization.

Solution — Question 4 (correct answer: B)

B is correct. The pharmaceutical timeline places Discovery, Preclinical Trials, and Clinical Trials (Phase 1–3) on the *expense (research)* side, with capitalization beginning around Regulatory Approval. The economic rationale is that until approval is obtained, technical feasibility of completion — the first of the six criteria — cannot be demonstrated, and the probability of failure remains material.

A is wrong because the label “trial” does not determine classification. Research encompasses the phase in which alternatives are still being evaluated; the boundary is the existence of a *chosen alternative*.

C is wrong and inverts the frameworks: US GAAP expenses all research and virtually all development, so it is the *less* permissive framework, not the more.

D is wrong because it describes retrospective capitalization based on hindsight. Recognition criteria are applied at the time expenditure is incurred; costs already expensed cannot be reinstated as assets.

Solution — Question 5 (correct answer: B)

B is correct. The amended treatment is precise: effective for fiscal years beginning after December 15, 2024, fair value measurement at each reporting date, with unrealized gains and losses recognized in earnings.

A is wrong because that is the *pre-2023* US GAAP treatment — indefinite-lived intangible assets at cost adjusted for impairment, with gains recognized only on sale. The exercise is to identify which regime applies to which period.

C is wrong because it describes the IFRS *revaluation model* asymmetry, not amended US GAAP. This distinction matters: as noted in section 15.7.4, the same holding can produce different net income under the two frameworks.

D is wrong because cryptocurrencies fail the definition of a monetary asset — money held or assets to be received in fixed or determinable amounts of money. Their volatility also disqualifies them from any cash-equivalent notion requiring insignificant risk of changes in value.

True or false?

Solutions — True/False

1 — FALSE. The critical part of the statement is the word *definition*. An internally generated brand may well satisfy the definition: it is non-monetary, has no physical substance, and is frequently separable — brands are bought and sold. What it fails is the **recognition criteria**, specifically that the cost be measurable reliably and, often, that future benefits be probable on evidence independent of management. The distinction is not pedantic: it explains why a *purchased* brand appears on the balance sheet while an identical internally generated one does not — the definition is satisfied in both cases, and only the recognition test separates them. Prada's own trademark, carried at €5.6 million against an external brand valuation of roughly USD 7.9 billion, is the illustration.

2 — FALSE. The statement contains a true premise and a false conclusion. It is correct that amortization and impairment produce no cash flow — Table 14.1 records “None” in the cash flow column for both. But the *capitalization decision itself* does affect the statement: it reclassifies the expenditure as a capex outflow, so the expenditure appears in **investing instead of operating**. CFO is therefore higher under capitalization. In the DELTA case, CFO was 3,360 under expensing and 6,360 under capitalization — a difference of 3,000 on identical cash movements. What is unaffected is *total* cash flow and free cash flow.

3 — FALSE. The critical word is *must*. Legal life functions as a **ceiling** on useful life, not as a determinant. The trademark example in section 15.4.5 makes this explicit: the asset has an unlimited legal life, but because the company expects to cease manufacture of the related product within 3 years, the useful life is **3 years** and the asset is amortized accordingly. Indefinite useful life must be positively justified by an expectation of indefinite benefit, and it carries consequences — no amortization, but mandatory annual impairment testing.

4 — FALSE. This overstates the finding. The evidence is that the AI/NI ratio has declined over 1972–2017 and that accounting earnings explain a **significant, but incomplete**, portion of stock price movements. The conclusion drawn is that financial statements — especially earnings — **are still useful, but they explain less of market value today compared to the past**. The correct analytical response is not to abandon financial statement analysis but to supplement it: to reconstruct the intangible investment the statements expense, and to attend to the non-financial indicators that the cellular-telephony evidence found value-relevant.

Exercise 14.4 — Reading Prada and Juventus like an analyst

Solution — Exercise 14.4

(a) Price-to-book for Prada.

The ratio is analytically unusable in its reported form. The denominator contains €867.9 million of intangibles reflecting *acquisition history* — principally the goodwill and trademarks arising from the purchases of Church's, Miu Miu, and related entities — while excluding the Prada brand, which the external valuation series places at roughly USD 7.9 billion in 2024. Book equity therefore omits the group's principal economic asset while including assets of limited standalone value.

The alternative anchor is **earnings power**. Since the brand manifests itself economically through pricing power and margin, the analyst should value the firm on sustainable operating margins and their persistence, cross-checked against the operating cash flow generated by those margins. The brand valuation series is useful as corroboration of trend and order of magnitude, but it must not be entered into the balance sheet: it is an external estimate, not an accounting measurement, and the evidence notes that capitalized brand assets exhibit *lower reliability* even when firms are permitted to recognize them.

(b) Tangible net worth and why the credit analyst computes it.

Tangible net worth = shareholders' equity – intangible assets. The intangible balance to be deducted is **€867.9 million**, of which €515.5 million is goodwill. Using the equity figure from the same consolidated statement of financial position — total shareholders' equity of €4,419.4 million at December 31, 2024 — tangible net worth is approximately $4,419.4 - 867.9 = €3,551.5$ million, about 80% of reported equity. The credit analyst deducts intangibles because:

- Goodwill has no separable existence and no recovery value in liquidation — it cannot be sold apart from the business it was created by acquiring;
- Store lease acquisition costs are effectively fully amortized (net €56 thousand against cost of €52.2 million), demonstrating that such assets are consumed rather than realizable;
- Trademarks, though separable in principle, are worth most to a buyer who also acquires the operating business — precisely the scenario that does not obtain in a distressed sale.

The consequence is that leverage covenants written on reported equity are materially more permissive than covenants written on tangible net worth, and a credit analyst must know which basis applies. Note the countervailing consideration: the *unrecognized* Prada brand is a genuine

source of cash flow that tangible net worth also excludes. Tangible net worth is therefore a measure of **liquidation protection**, not of **going-concern capacity**, and should be used accordingly rather than as a general measure of financial strength.

(c) The Juventus amortization ratio.

Accumulated amortization and write-downs of €268.8 million against historical cost of €573.3 million gives a ratio of approximately **47%**. Roughly half the cost of the first-team squad has been written off, implying that the squad is on average near the midpoint of its contractual life.

The risk this creates is a **reinvestment obligation**. Because amortization is driven by contract term rather than by economic value, the carrying amount declines mechanically toward zero as contracts run down. Unless the club continually acquires new registrations, the asset base — and with it the competitive capacity that generates revenue — erodes. The player schedule makes the timing explicit through the disclosed contract expiry dates, several falling on 30/06/26. The analyst should therefore read the amortization charge not merely as an accounting expense but as an estimate of the **recurring capital expenditure required to maintain the business**, and should test whether operating cash flow is sufficient to fund it.

(d) Reconciling measurement quality with predictive failure.

Juventus resolves every measurement obstacle that defeats brand accounting. Cost is objectively established by an arm's-length transfer fee. Useful life is contractually fixed and disclosed player by player. Identifiability is unambiguous: registration rights arise from legal rights and are separable — indeed they are routinely traded. Amortization is mechanical. By the standards of the recognition criteria of section 15.3.3, this is close to an ideal intangible asset.

And yet the academic evidence reports that while these assets are associated with current sales and operating profits, **they are not associated with these metrics for subsequent years**. The tension is concrete: a single player's market value rises from near zero to €45.0 million over eighteen months while the corresponding carrying amount can only decline.

The reconciliation is that **reliability and relevance are independent properties**. A reliably measured historical cost tells the analyst what was *paid*; it does not tell them what will be *earned*. The transfer fee capitalizes the market's expectation at the transaction date, and that expectation is immediately superseded by subsequent performance, injury, form, and market movement — none of which the carrying amount tracks. Amortization schedules reflect contractual duration, not economic decay, and the two diverge in both directions.

The general lesson, which extends well beyond football, is that solving the measurement problem does not solve the **prediction** problem. This is the strongest argument in the chapter that the limitations of intangible accounting are conceptual rather than technical: even where every practical measurement obstacle is removed, the resulting asset remains a record of past expenditure rather than a forecast of future benefit. For the analyst, this means that no accounting reform will remove the need to assess intangible value analytically, from earnings persistence and operating evidence, rather than reading it from the balance sheet.

Chapter 15 — The Influence of Accounting Standards on Financial Statement Analysis

Numerical Exercises

Exercise 15.1 — Heterogeneous Composition of an Account and Ratio Adaptation

(a) Relevance of intangibles. ALPHA: intangibles = 18,000 + 9,000 = 27,000; / total assets = 27,000 / 60,000 = **45.0%**; / net equity = 27,000 / 12,000 = **225.0%**. BETA: intangibles = 1,000 + 3,000 = 4,000; / total assets = 4,000 / 40,000 = **10.0%**; / net equity = 4,000 / 15,000 = **26.7%**.

(b) Unadjusted profitability. ALPHA: EBITDA / TA = 8,000 / 60,000 = **13.3%**; EBIT / TA = 3,500 / 60,000 = **5.8%**. BETA: EBITDA / TA = 7,000 / 40,000 = **17.5%**; EBIT / TA = 3,000 / 40,000 = **7.5%**. On both unadjusted measures BETA appears the more profitable company.

(c) Adjusted profitability. ALPHA: "Other" assets = 60,000 – 27,000 = 33,000; EBITDA / "Other" assets = 8,000 / 33,000 = **24.2%**. BETA: "Other" assets = 40,000 – 4,000 = 36,000; EBITDA / "Other" assets = 7,000 / 36,000 = **19.4%**.

(d) The ranking reverses. Unadjusted, BETA leads by 4.2 percentage points on EBITDA / total assets; adjusted, ALPHA leads by 4.8 percentage points. The reversal is entirely mechanical: ALPHA's denominator was inflated by an intangible balance that arose from the *history of its acquisitions*, not from the productivity of its network. Once the intangibles are removed from the denominator — and once the amortisation they generate is excluded from the numerator by using EBITDA — ALPHA's recurring asset base (current assets plus tangible fixed assets) proves to be the more productive of the two. The analytical conclusion is that **the unadjusted comparison was measuring acquisition history, not operating performance**.

(e) What the adjusted ratio does not measure. It is not a return on invested capital and must never be compared with a cost of capital. It deliberately excludes the capital that ALPHA actually committed — the cash paid for the acquisitions, which is now represented by the 27,000 of intangibles. Measured against the capital genuinely employed, ALPHA is the *less* profitable company: EBIT / total assets of 5.8% against 7.5%. Both statements are true and they answer different questions. ALPHA runs its physical network more productively; ALPHA has not yet earned an adequate return on the price it paid to acquire market position.

Tangible net worth. ALPHA: 12,000 – 27,000 = **–15,000**. BETA: 15,000 – 4,000 = **+11,000**. For a credit analyst this is the decisive number in the exercise. A goodwill impairment of 6,000 — one third of ALPHA's goodwill — would halve its reported equity and would double its debt-to-equity ratio without consuming a single unit of cash.

Exercise 15.2 — Impairment of a Long-Lived Asset: IFRS and US GAAP

(a) Impairment losses.

Scenario 1. US GAAP step 1: carrying amount 500 < undiscounted cash flows 540, so the recoverability test is passed and **no impairment loss is recognised** (the entity should, however, review the useful life and depreciation method). IFRS: recoverable amount = $\max(380 ; 420) = 420 < 500$, therefore **impairment = 80**.

Scenario 2. US GAAP step 1: $500 > 460$, the test is failed; step 2 measures the loss against fair value: $500 - 350 = 150$. IFRS: recoverable amount = $\max(350 ; 420) = 420$, therefore **impairment = 80**.

Scenario 3. US GAAP: $500 > 460$, test failed; loss = $500 - 440 = 60$. IFRS: recoverable amount = $\max(440 ; 420) = 440$, therefore **impairment = 60**. The two frameworks converge whenever fair value less costs of disposal exceeds value in use, because in that case both measure against the same benchmark.

(b) Neither framework is systematically more conservative. In Scenario 1, IFRS recognises 80 and US GAAP nothing, because the US recognition screen uses *undiscounted* cash flows, which are easier to clear. In Scenario 2, US GAAP recognises 150 and IFRS 80, because once the screen has been failed US GAAP measures against fair value alone and is not permitted to use value in use, whereas IFRS takes the **higher** of the two benchmarks. Which framework writes down more depends on the relationship between undiscounted cash flows, discounted cash flows and fair value for the specific asset.

(c) Variability. Compare Scenarios 1 and 2: the only change is a deterioration in undiscounted cash flows from 540 to 460 — some 15%. Under US GAAP the charge moves from 0 to 150; under IFRS it does not move at all (80 in both). The two-step architecture creates a **cliff**: nothing is recognised until the screen is failed, at which point the entire gap between carrying amount and fair value is recognised at once. The one-step architecture recognises deterioration continuously. A US GAAP earnings series through a declining cycle is therefore smoother and then discontinuous; an IFRS series is noisier but without the discontinuity. For trend analysis this matters more than the level of the charge, because a smooth series that ends in a cliff was never informative about the underlying deterioration.

(d) Following year's depreciation, Scenario 2. US GAAP: new carrying amount 350 over 5 years = **70 per year**. IFRS: new carrying amount 420 over 5 years = **84 per year**. The framework that recognised the larger loss now reports the *lower* future depreciation and therefore the *higher* future operating income. An impairment is, in this precise sense, a transfer of profit from the current period to future periods. The analyst who removes the impairment as an Unusual Item (Chapter 2) but leaves the subsequent depreciation unadjusted is double-counting the benefit.

(e) Reversal. Under IAS 36 an impairment loss on an asset other than goodwill may be reversed when the estimates used to determine the recoverable amount have changed — subject to the ceiling of the carrying amount that would have been determined, net of depreciation, had no

impairment been recognised. Continuing Scenario 2, as in part (d): the IFRS carrying amount one year later is $420 - 84 = 336$, and the ceiling is the carrying amount that would have been determined had no impairment occurred, $500 - (500/5) = 400$. Because the recoverable amount of 480 exceeds this ceiling, the reversal is capped: reversal = $400 - 336 = 64$, and the asset cannot be written up to 480. Under US GAAP the restoration of a previously recognised impairment loss on an asset held for use is **prohibited**. No reversal is recognised, and the carrying amount remains $350 - 70 = 280$. The IFRS series therefore moves in both directions with the cycle, while the US GAAP series records only the deterioration; over a full cycle the two carrying amounts diverge permanently.

Exercise 15.3 — Analytical Capitalisation of Operating Leases

(a) Present value and adjusted balance sheet. The annuity factor for five years at 8% is $[1 - 1.08^{-5}] / 0.08 = 3.9927$, so $PV = 250 \times 3.9927 = 998$. A right-of-use asset of 998 and a lease obligation of 998 are recognised. Adjusted total assets = $5,000 + 998 = 5,998$; adjusted fixed assets = $3,200 + 998 = 4,198$; adjusted total liabilities = $4,000 + 998 = 4,998$; owners' equity is unchanged at **1,000** because the lease is assumed to be newly commenced, so the asset and the liability are equal at the adjustment date.

(b) Adjusted income statement. The rental of 250 is removed from operating costs and replaced by depreciation of the right-of-use asset and interest on the lease obligation. $EBITDA = 600 + 250 = 850$. Depreciation = $300 + (998 / 5) = 300 + 199.6 = 499.6$. Adjusted EBIT = $850 - 499.6 = 350.4$. Interest = $200 + (998 \times 8\%) = 200 + 79.8 = 279.8$. Adjusted EBT = $350.4 - 279.8 = 70.6$, against 100 as reported. The reduction of 29.4 is the **front-loading effect**: in the early years of a capitalised lease, depreciation plus interest exceeds the straight-line rental it replaced.

(c) Ratios.

Debt to equity: $4,000 / 1,000 = 4.00\times \rightarrow 4,998 / 1,000 = 5.00\times (+25\%)$. Owners' equity / fixed assets: $1,000 / 3,200 = 31.3\% \rightarrow 1,000 / 4,198 = 23.8\%$. ROI (EBIT / total assets): $300 / 5,000 = 6.00\% \rightarrow 350.4 / 5,998 = 5.84\%$. EBITDA / interest expense: $600 / 200 = 3.00\times \rightarrow 850 / 279.8 = 3.04\times$. ROE (EBT / equity, taxes ignored): $100 / 1,000 = 10.0\% \rightarrow 70.6 / 1,000 = 7.06\%$.

(d) The four families do not move together, and this is the lesson of the exercise. The **capital ratios** deteriorate sharply — leverage rises by a quarter and the coverage of fixed assets by equity falls by more than seven percentage points — because the adjustment adds a liability of 998 against equity of only 1,000. The **coverage ratio** barely moves ($3.00\times \rightarrow 3.04\times$), because both the numerator and the denominator increase: the rental has been removed from EBITDA and part of it has reappeared as interest. **ROI falls slightly** ($6.00\% \rightarrow 5.84\%$): the numerator gains 50.4 while the denominator gains 998, and the second effect dominates. **ROE falls by 29%**, entirely through the front-loading of the expense. An analyst whose decision depends on leverage will reach a very different conclusion after the adjustment; an analyst whose decision depends on EBITDA coverage will reach almost the same one. Knowing *which* ratio the decision turns on determines whether the adjustment is worth performing at all — question 1 of Table 15.2.

(e) The age assumption. Equity was unaffected here only because the lease was assumed to be new, so the right-of-use asset equals the lease obligation. For a lease at mid-life the two amounts diverge: the asset is depreciated on a straight-line basis while the liability amortises on an effective-interest basis, which reduces the liability slowly at first. The asset therefore falls below the liability, and the difference is a **reduction of equity** in the adjusted balance sheet. Applied to a fleet of mid-life aircraft this effect is large, and it is the main reason why the adjusted debt-to-equity ratios in Table 15.16 reach 25× and 43×: the adjustment increases the numerator *and* reduces the denominator. The analyst who assumes new assets systematically understates the impact of the adjustment.

Exercise 15.4 — Gift Cards and Breakage under IFRS 15

(a) Recognition at 8% expected breakage. Expected breakage = $50,000 \times 8\% = 4,000$. Expected value to be used = $50,000 - 4,000 = 46,000$. Proportion of rights exercised = $18,400 / 46,000 = 40.0\%$. Product sales = **€18,400**. Breakage revenue = $40.0\% \times 4,000 = \text{€1,600}$. Total revenue = **€20,000**, which is also $40.0\% \times 50,000$ — the compact form of the rule. Residual contract liability = $50,000 - 20,000 = \text{€30,000}$.

(b) Recognition at 12% expected breakage. Expected breakage = 6,000; expected value to be used = 44,000; proportion of rights exercised = $18,400 / 44,000 = 41.82\%$. Product sales remain **€18,400**; breakage revenue = $41.82\% \times 6,000 = \text{€2,509}$. Total revenue = **€20,909**, and the residual contract liability falls to €29,091.

(c) What part (b) demonstrates. A four-percentage-point revision to an estimate produced €909 of additional revenue — an increase of 4.5% — **without a single additional item being sold**. The revision is a matter of judgement supported by historical data that only management holds. IFRS 15 codified the *method* and removed the classification chaos documented in Table 15.20, but it did not remove the estimate: this is the discretion dimension of Table 15.3 surviving codification. The equity analyst's response is therefore to read the disclosed breakage rate, to track its movement year on year, and to be alert to upward revisions in periods in which the company narrowly met a revenue target. A rising breakage rate is also, incidentally, an operating signal in its own right — it means a growing proportion of customers are not returning to the store.

(d) Before IFRS 15. The two competitors would both have recognised a liability on the initial sale, but the presentation of the breakage could have differed on two dimensions simultaneously. On the balance sheet, one might have shown the obligation as a separate gift card line item (11% of the surveyed companies) and the other within 'accrued expenses or other liabilities' (65%). In the income statement, one might have recognised breakage in net sales (64%) and the other in other income (21%) — so that the same event increases *Core Operating Income* for one company and *Non-Core Operating Income* for the other, in the terminology of Chapter 2. Since only 5% of the surveyed companies disclosed the amount of breakage, the analyst could generally neither reclassify nor quantify: the correct professional response was Level 3 of Table 15.1 — proceed, and state the limitation.

Multiple-Choice Questions

MCQ 1

Correct answer: C. This is Level 3 of the three-level response in Table 15.1. When adjustment is not feasible, the limitation becomes part of the analytical output: it qualifies the conclusion and sizes the residual uncertainty for the decision-maker.

A is wrong because compliance and comparability are different properties. The analyst has already established that the numbers do not adequately represent operations *for the specific analytical objective*; an unqualified audit opinion says nothing about that.

B is wrong on two counts. It substitutes an unverifiable estimate for a disclosed number, and — more seriously — it presents the result without flagging that it is an estimate, which removes the reader's ability to assess the reliability of the conclusion. Estimating is legitimate; presenting an estimate as a reported figure is not.

D is wrong because it discards information. Time-series analysis of each company is useful but does not answer the cross-sectional question the analyst was asked. The correct response preserves the comparison and attaches the caveat.

MCQ 2

Correct answer: B. The problem is one of *classification*, not of measurement, and the disclosure permits the item to be located. A reformulation changes where the amount sits without changing the amount — response 2 in Table 15.4, and exactly the technique developed in Chapter 2.

A is wrong because it is disproportionate. The measurement is not in dispute; an accounting adjustment would introduce the analyst's own assumptions into numbers that do not require them, making the analysis less reproducible for no gain.

C is wrong because a classification difference does not average out over time. If one company reports the item within operating income every year and the other below it every year, five years of data reproduce the distortion five times.

D is wrong because the bottom line is not what the analyst is comparing. Operating income, EBITDA, the operating margin and every ratio built on them are all affected, and the ROI denominator must be coherent with the numerator — the point established in Chapter 2.

MCQ 3

Correct answer: C. As Table 15.11 and Exercise 15.2 demonstrate, IFRS recognises a loss where US GAAP recognises none when undiscounted cash flows clear the carrying amount but value in use does not; conversely US GAAP recognises a larger loss than IFRS once the recoverability screen has been failed and fair value is below value in use.

A is wrong because it generalises a single scenario. It is true that the US recognition *screen* is easier to clear; it does not follow that the US *measurement* is less severe, and in Scenario 2 of Exercise 15.2 US GAAP writes down 150 against IFRS's 80.

B is wrong for the mirror reason: it generalises the measurement step and ignores the recognition step, in which US GAAP may recognise nothing at all.

D is wrong because the benchmarks themselves differ — fair value under US GAAP, the higher of fair value less costs of disposal and value in use under IFRS — so the amounts coincide only in the special case where fair value exceeds value in use.

MCQ 4

Correct answer: B. The postponement of investments to increase cash flow appears explicitly in the left-hand column of Table 15.12. The substance of the business has genuinely changed — the expenditure has not been incurred — and the accounts report that change faithfully.

A is wrong because nothing has been misrepresented. In abuse, the *form* of a transaction is distorted so that the accounts report something other than the substance; here there is no transaction at all, and the accounts say so.

C is wrong because a transaction that did not occur cannot be reclassified. There is no amount to move. The analyst's response is not a reformulation but a *behavioural* diagnosis, performed on the multi-year pattern of capital expenditure relative to depreciation and relative to peers.

D is wrong because there is no accounting practice in question. A non-codified practice arises where a transaction has occurred and the standards have not yet specified how to account for it — the gift card case of Section 15.7.

A note on severity. The fact that B is not an accounting violation does not make it harmless. Real earnings management is legal, invisible in the statements, and frequently more destructive of long-run value than the manipulations that attract enforcement action — which is precisely why the distinction between the two categories is described as 'tentative'.

MCQ 5

Correct answer: B. The ratio is a *consistency adjustment* — response 3 in Table 15.4. Its output is a benchmark: operating profitability on the most recurring assets of approximately 30% for the European telecommunications peer group.

A is wrong, and it is the most dangerous of the distractors. The ratio is measured before depreciation and excludes from the denominator the capital actually invested in acquisitions; comparing it with a cost of capital would systematically favour the most acquisitive companies in the group — the opposite of the correction being attempted.

C is wrong: nothing in the construction values the intangibles. Their carrying amount is simply removed from both sides.

D is wrong on a point of fact. IFRS governs the preparation of financial statements, not the ratios an analyst chooses to compute from them; EBIT / total assets remains perfectly computable and is reported in Table 15.8 alongside the adjusted measure.

True / False

T/F 1

FALSE. The critical word is *always*. The undiscounted screen makes the US GAAP recognition threshold easier to clear, so there are cases in which US GAAP recognises nothing while IFRS recognises a loss. But once the screen has been failed, US GAAP measures the loss against **fair value**, while IFRS measures it against the **higher** of fair value less costs of disposal and value in use. Whenever value in use exceeds fair value — the normal case for an asset that is worth more in the entity's business than in the market — the US GAAP loss is the larger of the two. Scenario 2 of Exercise 15.2 gives 150 under US GAAP against 80 under IFRS.

T/F 2

TRUE. This is real earnings management — a physiological behaviour in the vocabulary of this chapter. The expenditure genuinely was not incurred; the reported cost saving and the reported cash saving are both real; and no accounting standard has been violated. The critical distinction is that the *substance* has changed, and the accounts faithfully report the changed substance.

The proposition is nonetheless a trap for the unwary reader, because 'technically correct' is not the same as 'analytically informative'. The reported earnings figure is now a poor guide to the company's recurring earnings capacity, and the damage — a depleted research pipeline — will appear in revenue several years later, in a period whose accounts will contain no trace of the decision that caused it. The analyst's instrument here is not an adjustment but a multi-year ratio of R&D to revenue, benchmarked against peers.

T/F 3

FALSE. The critical clause is *compared with the weighted average cost of capital*. A comparison with the cost of capital requires a return measured on the whole of the capital invested, after the depreciation of that capital, and on an after-tax basis — the NOPAT / Invested Capital construction of Chapter 6 and the value-added logic of Chapter 7. The adjusted ratio of this chapter fails all three conditions: it is before depreciation and amortisation, before tax, and its denominator deliberately excludes the intangible assets that represent capital the acquisitive companies genuinely paid for. It is a peer-group comparability instrument, and its output is a ranking and a benchmark, not a value judgement.

T/F 4

FALSE. The critical word is *any*. It is true that the specific abuse described in this chapter — the recurring use of operating lease accounting to keep financial liabilities off the balance sheet — has been closed for lessees of any size, and that the *balance sheet* comparison is now broadly homogeneous. But the income statement is not. IFRS 16 applies a single lessee model in which every capitalised lease produces depreciation plus interest, whereas ASC 842 retains a dual classification under which leases classified as operating continue to produce a single straight-line lease cost inside operating expenses. **The consequence is that EBITDA is systematically higher for the IFRS reporter** with an identical lease portfolio, because the lease cost has been moved below the EBITDA line. Any EV/EBITDA or leverage-to-EBITDA comparison across the two frameworks still requires an adjustment; what has changed is its nature, from capitalising a liability to normalising an expense classification.

T/F 5

FALSE. The critical word is *removed*. IFRS 15 codified the **method**: a contract liability on issue, and recognition of the expected breakage amount as revenue in proportion to the pattern of rights exercised by customers. This eliminated the classification dispersion documented in Panels B and C of the pre-IFRS 15 survey, where the same event was recognised in net sales by 64% of companies, in other income by 21%, and as a reduction of cost of goods sold or of SG&A by others.

What the standard did not eliminate is the **estimate**. The expected breakage rate is a managerial judgement based on historical redemption data held only by the company, and it feeds revenue directly: Exercise 15.4 shows that revising the rate from 8% to 12% increases recognised revenue by 4.5% with no incremental sales. Codification narrows the space in which discretion operates; it does not close it.

T/F 6

TRUE. Microsoft announced the change on 8 July 2003, with effect from September 2003, and simultaneously undertook to expense equity compensation on a voluntary basis with retrospective restatement of prior years — which is what the 'revised' lines of Table 15.18 report. FASB Statement No. 123(R) was issued on 16 December 2004 and applied to periods beginning after 15 June 2005.

The analytical lesson is stronger than it first appears. **The anticipation of a change in accounting standards was sufficient to change corporate behaviour before the standard existed.** Managers do not respond only to rules in force; they respond to rules they expect to come into force, which means that the effects of a standard on real corporate decisions can precede its effective date by years — a point that matters whenever the analyst is examining a transition period.

Real-World Interpretation Exercise

Exercise 15.5 — A Credit Note on the European Telecommunications Peer Group

Model answer, paragraph 1 — the governing feature. The credit assessment of this peer group is governed by the relationship between intangible assets and net equity. For three of the five operators, intangibles exceed book equity: 212.5% for Telefónica, 186.3% for Telecom Italia and 158.8% for France Télécom; Vodafone (81.0%) and América Móvil (58.2%) retain positive tangible net worth. Tangible net worth is therefore negative for three of the five operators and thin for the group as a whole, and an asset-cover test would be breached at inception by three borrowers. The practical consequence is that the facility cannot be underwritten on the value of the balance sheet; it must be underwritten on the recurring cash generation of the network. Table 15.8 supports this reading: once intangibles are removed from both sides of the ratio, four of the five operators generate EBITDA equal to 29–33% of their recurring asset base, which is a stable and defensible industry benchmark.

Model answer, paragraph 2 — ratios, covenants and the borrower of concern. The reported EBIT-based ratios must not be used to rank these borrowers. EBIT / total assets ranges from +14.7% to –13.0%, but the dispersion is created after EBITDA by amortisation and impairment charges arising from intangible balances that differ by a factor of nearly three as a proportion of total assets — that is, by acquisition history rather than by operating performance. EBITDA margins, by contrast, cluster between 25.5% and 41.1%, consistent with the industry's cost structure. The committee should therefore require covenants built on cash-flow measures — net debt to EBITDA and EBITDA to interest expense — with an explicit contractual definition of EBITDA that excludes impairment, rather than a net-worth or asset-cover test that a non-cash write-down could breach in a year of unimpaired cash generation. The borrower carrying the highest accounting-driven risk is **Telecom Italia**: goodwill alone represents 161.4% of equity and 48.9% of total assets, so an impairment of one fifth of goodwill would eliminate approximately one third of book equity without any deterioration in operating cash flow. Vodafone illustrates that this is not a theoretical exposure — its 40,916 goodwill impairment converted an EBITDA of 13,042 into an EBT of –44,949 in a single year.

Marking guidance. Full credit requires the candidate to (i) reach the tangible-net-worth conclusion from the equity ratios rather than asserting it; (ii) distinguish explicitly between the EBITDA convergence and the EBIT dispersion, and attribute the latter to accounting rather than to operations; (iii) derive the covenant structure *from* the balance-sheet finding rather than listing covenants generically; and (iv) support the choice of borrower with a quantified impairment sensitivity. A candidate who concludes that Vodafone is the weakest credit on the basis of its reported EBIT of –13.0% has failed the central point of the chapter.

Chapter 16 — Accounting for (Dis)Value. Misreporting, Fraud, and the Institutions of Reporting Quality

18.10.1 Numerical Exercises

Exercise 16.1 — Detection Is Not Incidence

(a) Observed rates. United States: $120 / 4,000 = 3.00\%$. Europe: $45 / 3,000 = 1.50\%$. On the observed figures, American companies appear to misreport twice as often as European ones.

(b) Implied incidence. Using $\text{observed} = \text{incidence} \times P(\text{detection and disclosure})$: United States, $120 / 0.75 = 160$ instances, or $160 / 4,000 = 4.00\%$ of companies. Europe, $45 / 0.40 = 112.5$ instances, or $112.5 / 3,000 = 3.75\%$ of companies.

(c) The comparison. The observed ratio is $3.00 / 1.50 = 2.00\times$. The implied ratio is $4.00 / 3.75 = 1.07\times$. Almost the whole of the apparent difference in corporate conduct is an artefact of the difference in detection and disclosure. Note also that the direction of the ranking is preserved but its magnitude collapses — the qualitative conclusion survives, the quantitative one does not.

(d) Equalising probability. For the implied incidences to be equal, Europe would need $4.00\% \times 3,000 = 120$ instances, so $45 / x = 120$, giving $x = 0.375$. A European detection-and-disclosure probability of 37.5% — only 2.5 percentage points below the assumed 40% — would make the two markets identical in underlying conduct. The conclusion is extremely sensitive to a parameter the analyst cannot observe.

(e) What is established. The exercise establishes that observed restatement frequencies are not interpretable as incidence without an assumption about detection, and that plausible assumptions overturn the naive comparison. It does not establish that the two markets are in fact equivalent: the detection probabilities are assumed, not measured, and the sensitivity computed in (d) is precisely the reason the analyst should report a range rather than a point. The correct professional output is a statement of the form: on any reasonable assumption about relative enforcement intensity, the difference in observed restatement rates cannot support an inference about relative reporting quality.

Common mistake. Dividing the two observed rates and reporting that misreporting is twice as common in the United States. This is the error identified in §16.4.3, and it is the single most frequent error in cross-border accounting-quality comparisons.

Exercise 16.2 — The Mechanics and the Consequences of a Restatement

(a) Restated figures. Revenue $1,200 - 60 = 1,140$. Cost of sales falls by 36, so operating income falls by $60 - 36 = 24$, to 126. Interest expense is unchanged at 30, so profit before tax is 96, tax at 30% is 28.8, and net income is 67.2. The reduction in net income is 16.8, which is $24 \times (1 - 0.30)$.

Balance sheet. Trade receivables fall by 60 to 240; inventory rises by 36 to 236; the current tax liability falls by 7.2; equity falls by 16.8 to 463.2. Total assets fall by $60 - 36 = 24$, to 1,476. Liabilities

fall by 7.2 and equity by 16.8, together 24. The balance sheet balances. Net financial debt is unchanged at 600: no cash has moved, which is the point — the restatement corrects a representation, not a transaction.

(b) Ratios. Return on sales (EBIT / revenue): $150 / 1,200 = 12.50\%$ before, $126 / 1,140 = 11.05\%$ after. Return on investment (EBIT / total assets): $150 / 1,500 = 10.00\%$ before, $126 / 1,476 = 8.54\%$ after. Return on equity (net income / equity): $84 / 480 = 17.50\%$ before, $67.2 / 463.2 = 14.51\%$ after. Debt to equity: $600 / 480 = 1.25$ before, $600 / 463.2 = 1.295$ after. Every profitability measure deteriorates and leverage rises: the numerator falls proportionally more than the denominator in each case, because the reversed margin (24 on 60, that is 40%) is far higher than the firm's average margin.

(c) Growth. Before: $1,200 / 1,100 - 1 = 9.09\%$. After: $1,140 / 1,100 - 1 = 3.64\%$.

(d) Which effect dominates. The level effects in (b) are material but bounded: return on investment falls by about 1.5 percentage points, leverage rises by 4%. The growth effect is of a different order: the growth rate falls by 60% of itself, from 9.09% to 3.64%. In any valuation in which the terminal value is a function of the growth rate — which is to say, in any valuation — the revision of growth from 9% to under 4% will dominate the revision of the current-year margin by an order of magnitude. This is the general lesson of the exercise: a restatement that removes revenue from one period does not merely reduce a level, it re-describes the trajectory, and the trajectory is what the market was capitalising.

Common mistakes. First, reversing revenue of 60 against net income of 60 and forgetting the cost of sales and the tax, which overstates the equity effect by a factor of more than three. Second, forgetting to return the goods to inventory, in which case the balance sheet does not balance. Third, recomputing the ratios and stopping there, without re-examining the growth rate or the revenue recognition policy that produced the error.

Exercise 16.3 — Pricing a Spillover

(a) Expected value effects. Average peer: $2,400 \times 0.5\% = €12.0\text{m}$. High-accrual peer: $2,400 \times 1.2\% = €28.8\text{m}$. High-accrual peer sharing the auditor: $2,400 \times 1.6\% = €38.4\text{m}$. Interlock evidence taken alone: $2,400 \times 1.03\% = €24.7\text{m}$. Note that these estimates are not additive across the two literatures — the interlock and the accounting-similarity channels are estimated on different samples and overlap — so the defensible statement is a range of roughly €12m to €38m rather than a sum.

(b) The two interpretations. The observed fall of 1.0% (€24m) lies between the average-peer and the high-accrual estimates. Either the market has not yet identified ZETA as a high-accrual firm sharing the offender's auditor — in which case there is further downside as the analysis becomes widespread, and the analyst who has done the work early holds genuinely private information — or the market has identified it and has offset it against something the analyst has not modelled, such as an expectation that ZETA gains market share from a distracted competitor. The two are distinguishable by looking at what the market did to other peers: if the high-accrual peers as a

group fell by roughly 1.2% and ZETA fell by 1.0%, the market is discriminating correctly and the residual is idiosyncratic. If the whole peer group fell uniformly by about 1.0%, the market is not discriminating and the analytical opportunity is real.

(c) The action. For the equity analyst the finding is a trading signal only if the accrual position is both correct and not yet priced; the more durable output is a re-examination of ZETA's own revenue recognition policy, since the restating competitor's error concerned the same practice in the same industry. For the credit analyst the share price is largely irrelevant: what matters is whether ZETA's receivables contain the same arrangement, whether the auditor — now under scrutiny elsewhere — will take a harder line at the next year-end, and whether the covenant headroom would survive a similar reversal. The credit response is to model the reversal explicitly, as in Exercise 16.2, and to test the covenants against the restated figures.

Common mistake. Adding the four percentages together to produce a single expected loss. The estimates come from separate studies with overlapping samples and different windows; summing them implies an independence that has not been established.

Exercise 16.4 — Reading a Discontinuity

(a) Expected frequencies. On the left, the two nearest uncontaminated intervals are $[-1.5\%, -1.0\%) = 262$ and $[-1.0\%, -0.5\%) = 250$, a decline of 12 per interval as one moves toward zero; extrapolating gives an expected 238 in $[-0.5\%, 0)$. On the right, the two nearest uncontaminated intervals are $[+1.0\%, +1.5\%) = 286$ and $[+0.5\%, +1.0\%) = 300$, an increase of 14 per interval as one moves toward zero; extrapolating gives an expected 314 in $[0, +0.5\%)$.

(b) Deficit and excess. Below zero: $238 - 168 = 70$ firms fewer than expected. Above zero: $402 - 314 = 88$ firms more than expected.

(c) Proportion. $70 / 238 = 29.4\%$ of the firms that would have been expected to report a small loss are absent from that interval. The order of magnitude is consistent with the published estimate that between 30% and 44% of firms with small pre-managed losses report a profit instead.

(d) Reconciliation. The deficit (70) and the excess (88) need not be equal, and here they are not. Three reasons: firms may cross from further below zero, so the excess draws on more than one interval of deficit; the linear extrapolation is an approximation of a curved density; and with counts in the hundreds, sampling variation of ± 10 to 15 firms is unremarkable. The professional treatment is to report both figures and to conclude that between roughly 70 and 90 firms in this universe are in the interval above zero for reasons unrelated to their economics — not to insist on a single number.

(e) What may be concluded about one firm. Almost nothing. The interval $[0, +0.5\%)$ contains 402 firms, of which the analysis suggests some 88 arrived there through discretion — meaning that roughly four out of five firms in that interval reported what they earned. A firm's presence in the interval raises the probability that its result is managed from a base rate of a few per cent to something near 20%, which is a reason to inspect the accruals of that firm and no reason at all to

conclude anything about it. The distribution generates a prior; only the firm's own reconciliation of earnings to cash flow can update it.

Common mistakes. Comparing the interval just above zero with the interval just below it (402 against 168) and reporting a ratio of 2.4 as the measure of earnings management; this conflates the discontinuity with the underlying slope of the distribution. And treating the individual firm as identified, which is the inferential error set out in §16.2.2.

18.10.2 Multiple-Choice Questions

MCQ 1

Correct answer: C. The evidence establishes materiality, an accounting trace and a motive, which is sufficient to downgrade the reliability of the reported figure and to re-specify the analysis. That is the professional output described in §16.1.1.

A is wrong because fraud requires intentionality, which no external evidence in this fact pattern establishes. Asserting it is both unsupported and, in a published report, actionable.

B is wrong on its reasoning even though its conclusion is plausible. A divergence between revenue growth and collections can arise from a genuine change in customer mix, in geography, or in contractual payment terms during an expansion. The observation is consistent with earnings management; it does not establish it.

D is wrong because it confuses two different conclusions. The analyst cannot conclude that management intended to mislead; the analyst can perfectly well conclude that the number is unreliable, which is the conclusion the decision actually requires. Treating unobservable intent as a bar to any judgement would make financial statement analysis impossible.

MCQ 2

Correct answer: C. A material prior-period error is corrected retrospectively, and the comparatives are restated. This is the accounting event that Table 16.4 identifies as a restatement in the strict sense.

A is wrong: a revision of a useful life is a change in accounting estimate, applied prospectively and disclosed. Prior periods are not touched, however large the effect.

B is wrong: this is a reclassification. The comparatives are re-presented so that the two years remain comparable, but no error is being corrected, and no total changes. The analytical work it generates is a re-run of the reformulation, not a rebuilding of the series.

D is wrong for the same reason as A: a provision updated for new information is a change in estimate. Note the trap the question sets — the effect on reported profit may be far larger than that of the error in C, and it is nevertheless not a restatement. Magnitude does not determine the accounting treatment; the nature of the event does.

MCQ 3

Correct answer: C. This is the decomposition of §16.4.3, quantified in Exercise 16.1. The American system raises the second and third terms deliberately, through auditor inspection, personal certification, an internal-control audit, whistleblower incentives and private litigation.

A is wrong: it treats the output of a detection system as a measure of the behaviour being detected — the same error as inferring higher crime from more police reports.

B is wrong although it is superficially sophisticated. It reverses the naive inference without escaping it: it still reads a single observed frequency as though it isolated one of the three terms.

D is wrong because it over-corrects. Restatement data are informative; they are simply not interpretable without an assumption about detection. The professional response is to state the assumption and test the sensitivity of the conclusion to it, as in Exercise 16.1(d) — not to discard the data.

MCQ 4

Correct answer: B. In the evidence of §16.7.1 the peer reaction is concentrated among firms with high industry-adjusted accruals, and the penalty is larger where the peer shares the restating firm's auditor. The market is re-assessing accounting quality along the dimension that failed, not marking down an industry.

A is wrong because leverage is not the dimension on which the market discriminates here. Leverage determines how much a given revision of value hurts the equity, but the contagion evidence concerns which firms are revised, not by how much a revision is amplified.

C is wrong because it confuses the euro loss with the percentage return. The abnormal return is measured in percentage terms; the largest firm has the most value at risk but is not thereby the most suspected.

D is wrong, and it is the answer that a naive information-transfer story would give. The whole analytical content of the finding is that the reaction is differentiated — which is what makes it usable by an analyst who can compute accruals before the market does.

MCQ 5

Correct answer: B. The direction of manipulation follows the incentive. Where the dominant incentive is fiscal rather than capital-market, the evidence on Italian private firms in §16.7.5 documents downward manipulation together with tax aggressiveness — and, in the same population, higher bank debt, lower cash holdings and a materially higher probability of default.

A is wrong because it treats conservatism as a synonym for quality. Deliberate understatement is a misrepresentation of the firm's economics in the same sense as overstatement, and it conceals capacity to service debt as effectively as overstatement conceals its absence.

C is wrong on the definition. Earnings management is the deliberate intervention in the reporting process to achieve a desired level of reported earnings; nothing in the definition specifies a direction.

D is wrong although its premise is sound. A lender does care principally about cash, but reported earnings drive the covenants, the tax position affects the cash actually available, and a firm willing to misrepresent in one direction has demonstrated the willingness and the technical capacity to misrepresent in the other when the incentive changes — for example, when it next needs to borrow.

18.10.3 True / False

True/False 1

FALSE. The critical word is any. Three of the four constituent elements of fraud — materiality, misrepresentation or omission, and the presence of a motive — leave observable traces, and the whole analytical apparatus of this book is directed at them. What the analyst cannot do is characterise the conduct as fraudulent, which is a judgement reserved to enforcement bodies and courts. The distinction is between the reliability of a number, which the analyst assesses, and the culpability of a person, which the analyst does not.

True/False 2

FALSE. The critical clause is the causal one — preferable because it remains within the standards. Compliance with the standards is a statement about legality, not about value. Accrual manipulation misleads investors while leaving the business intact, and reverses mechanically in later periods; real manipulation reports the truth about a business that management has deliberately damaged, and never reverses. A cancelled research programme, a postponed maintenance cycle or a discount granted to pull sales forward destroys value permanently and is accounted for correctly at every step. The adjective physiological used in Chapter 15 describes the legality of the act, not its consequences.

True/False 3

FALSE. The critical word is likely, applied to an individual firm. The distributional evidence is a statement about a population: it establishes that the interval immediately above zero contains more firms than the shape of the distribution can explain, and Exercise 16.4 shows that the excess is on the order of one firm in five in that interval. Four out of five firms in the interval reported what they earned. The finding raises the prior probability enough to justify inspecting that firm's working-capital accruals; it does not make manipulation likely for that firm.

True/False 4

FALSE. The critical word is only. A restatement destroys the historical series on which every trend, growth rate and average margin was estimated, so it affects the analysis of periods that are themselves unchanged. It is informative about accounts that were not restated, because the same

system of estimation and the same judgement produced them. And it raises the probability of subsequent events — executive turnover, auditor change, covenant renegotiation, litigation — that alter the distribution of future outcomes. Correcting the level without revising the trajectory and the risk is half the work, as Exercise 16.2(d) demonstrates numerically.

True/False 5

FALSE. The critical element is the chronology. Directive 2014/56/EU and Regulation (EU) No 537/2014 were adopted in 2014 and applied from June 2016; Wirecard collapsed in June 2020. The European audit reform was the delayed response to the financial crisis and to the earlier generation of European accounting failures. The genuine post-Wirecard reform was national and later: Germany's Act on Strengthening Financial Market Integrity, in force from 1 July 2021.

Note that the underlying proposition of §16.5 — that regulation follows scandal rather than preventing it — is not weakened by this correction but reinforced by it: a reform adopted in 2014 did not prevent the 2020 failure, and the response to that failure arrived in 2021.

True/False 6

TRUE. This is the contagion result of §16.7.3: the indicator for a board link to an identified manipulator is positive and significant at the 5% level across specifications, and the same architecture appears in the diffusion of option backdating, where the association is with links to identified backdaters specifically rather than with board links in general.

The critical qualification, which a complete answer should state, is that the finding is about propensity in a population and not about any individual board. It supports extending an investigation to connected firms; it does not by itself impugn a director.

18.10.4 Real-World Interpretation Exercise

Exercise 16.5 — A Note for the Credit Committee

Model answer, paragraph 1 — what is established. Nothing has been established about the borrower's own accounts, which are unchanged, unqualified and not under review. What has changed is the probability that they contain the same problem, and it has changed through three independent channels. The competitor's error concerned revenue recognition in the same industry, so the practice at issue is one our borrower may also apply; the two companies share an external auditor, and the evidence on restatement contagion shows the market penalty concentrating precisely on peers sharing the offender's auditor and carrying high industry-adjusted accruals; and the shared director chairs our borrower's audit committee, the position the interlock evidence identifies as carrying the largest reputational penalty because it is the function most directly implicated. Against this, the borrower reports in a jurisdiction in which corrections are disclosed quietly and late, so the absence of any announcement to date carries little information. The honest summary is that our estimate of the reliability of the reported interest coverage of 3.2× has widened, in one direction only, while our estimate of the borrower's operations is unchanged.

Model answer, paragraph 2 — what the bank should do. Three actions, none of which presupposes wrongdoing. First, model the restatement: apply the competitor's disclosed error to our borrower's own revenue base and test whether coverage would survive; on the competitor's proportions a reversal of that magnitude would take coverage below the 3.0× covenant, which converts an accounting question into a credit question and justifies the work. Second, exercise the information rights we already hold rather than seeking new ones: request the ageing of receivables, the contractual terms governing rights of return, and the auditor's report to the audit committee on revenue recognition — all documents the borrower can supply without any implication of suspicion. Third, do not accelerate, do not reprice on this evidence, and do not communicate a view of the borrower's conduct: we have a change in the variance of our estimate, not in its central value, and the appropriate response to variance is information and, if the facility is renewed, a covenant on receivables days — not enforcement. Reassess when the requested documents arrive or at the half-year, whichever is earlier.

What the exercise tests. The ability to hold two propositions simultaneously — that nothing is proven and that something material has changed — and to convert the second into action without asserting the first. An answer that recommends acceleration fails on evidence; an answer that recommends no action fails on diligence.

Chapter 17 — Forecasting Financial Statements and Firm Valuation: From Pro Forma Statements to Value per Share

Exercise 17.1 — Building an articulated one-year forecast (income, equity and cash flow)

(a) *Projected income statement (€ m).* Revenue $500.0 \times 1.10 = 550.0$. Gross margin in Year 0 is $(500 - 300)/500 = 40\%$, so cost of goods sold $= 60\% \times 550.0 = 330.0$ and gross profit $= 220.0$. SG&A $= 20\% \times 550.0 = 110.0$. Depreciation $= 12\% \times 250.0 = 30.0$. Operating income $= 220.0 - 110.0 - 30.0 = 80.0$. Debt is 50% of Year 0 total assets (200/400), and Year 1 total assets before the plug are 440.0 (part c), so Year 1 debt $= 220.0$ and interest $= 6\% \times (200.0 + 220.0)/2 = 12.6$. Income before taxes $= 67.4$; taxes at 25% $= 16.85$; net income $= 50.55$. Dividends $= 30\% \times 50.55 = 15.165$; retained earnings $= 35.385$.

(b) *Implied capital expenditure.* $275.0 - 250.0 + 30.0 = 55.0$ — that is, 1.83 times depreciation. The firm is expanding its asset base in line with the 10% growth assumption.

(c) *Projected balance sheet (€ m).* Cash $3\% \times 550.0 = 16.5$; receivables $12\% \times 550.0 = 66.0$; inventories $25\% \times 330.0 = 82.5$; net PP&E 275.0; **total assets before plug 440.0**. Payables $16.67\% \times 330.0 = 55.0$; other current liabilities $4\% \times 550.0 = 22.0$; debt 220.0; equity $130.0 + 35.385 = 165.385$; **total liabilities and equity 462.385**. Liabilities and equity exceed assets by **22.385**: a **financing surplus**. Under the stated policy it is invested in marketable securities, bringing total assets to 462.385.

(d) *Projected statement of cash flows (€ m).* Net income 50.55 + depreciation 30.0 – increase in receivables 6.0 – increase in inventories 7.5 + increase in payables 5.0 + increase in other current liabilities 2.0 = **operating cash flow 74.05**. Investing: capital expenditure (55.0) and purchase of marketable securities (22.385) = **(77.385)**. Financing: net debt increase 20.0 less dividends 15.165 = **4.835**. Net change in cash $= 74.05 - 77.385 + 4.835 = 1.5$, which equals $16.5 - 15.0$. **The model articulates.**

(e) *Free cash flow for common equity.* $74.05 - 1.5$ (increase in the operating cash requirement) – 55.0 (capital expenditure) + 20.0 (net debt) – 22.385 (into financial assets) = **15.165**, exactly equal to dividends declared.

Economic meaning. ALFA converts net income of 50.55 into free cash flow to equity of 15.165 — a conversion rate of 30% — because 55.0 of capital expenditure and 8.0 of net working capital investment stand between the two. The equality of free cash flow to equity and dividends is not a result about ALFA; it is a property of any fully articulated forecast, and it is why the dividends and free-cash-flow valuation models cannot disagree (§17.7.4).

Common mistakes. (i) Treating capital expenditure as an assumption rather than deriving it from the roll-forward. (ii) Computing interest on the closing rather than the average debt balance, which overstates interest expense by 0.6 here. (iii) Forcing the balance sheet to balance by adjusting an operating account — most often inventories — which destroys the turnover assumption and produces a model that no longer describes the business. (iv) Forgetting that the

increase in the *operating* cash requirement is a use of funds in the free cash flow ladder, while the marketable securities balance is a financing flow, not an operating one.

Exercise 17.2 — Cost of capital, tax rates and weights

(a) Weighted average yield = $(400 \times 3.0\% + 600 \times 5.0\%) / 1,000 = 4.20\%$. After tax at the statutory rate: $4.20\% \times (1 - 0.24) = 3.192\%$. At the effective rate: $4.20\% \times (1 - 0.15) = 3.570\%$.

(b) Cost of equity = $3.2\% + (1.15 \times 5.0\%) = 8.95\%$.

(c) Market-value weights: debt $1,000/4,000 = 25\%$; equity $3,000/4,000 = 75\%$. WACC with the statutory rate = $0.25 \times 3.192\% + 0.75 \times 8.95\% = 7.5105\%$; with the effective rate = **7.6050%**. Valuing the operating assets as a growing perpetuity: $250 / (0.075105 - 0.02) = \text{€}4,536.8\text{m}$ versus $250 / (0.07605 - 0.02) = \text{€}4,460.3\text{m}$. The 9.5 basis point difference in the rate produces a **1.7%** difference in value.

(d) With book equity of 800, the weights become debt $1,000/1,800 = 55.6\%$ and equity 44.4% . WACC = $0.556 \times 3.192\% + 0.444 \times 8.95\% = 5.751\%$, and the value becomes $250 / (0.05751 - 0.02) = \text{€}6,664.7\text{m}$ — **47% higher**. The error arises because book weights overstate the share of the cheapest source of capital.

Economic meaning. The tax-rate question is second-order for a firm with modest leverage but first-order for a leveraged one, since the tax shield is multiplied by the debt weight. The weighting question is first-order always. This is why the definition in §17.5.6 specifies market values, and why the circularity discussed in §17.5.6 and §17.7.5 — market equity weights in a model whose purpose is to test the market value of equity — has no fully satisfactory solution other than iteration or the use of stated target weights.

Common mistakes. (i) Using the coupon rate rather than the yield to maturity. (ii) Applying the tax shield to the cost of *equity* as well as to debt. (iii) Using book values because they are the numbers on the balance sheet — the single most common and most costly error in cost-of-capital estimation. (iv) Failing to notice that as $(\text{WACC} - g)$ narrows, small differences in the rate produce large differences in value.

Exercise 17.3 — The equivalence of the dividends and residual income models, and the T+1 trap

(a) Using $D_t = CI_t + BV_{t-1} - BV_t$: $D_1 = 120.0 + 1,000.0 - 1,040.0 = 80.0$; $D_2 = 132.0 + 1,040.0 - 1,100.0 = 72.0$; $D_3 = 145.2 + 1,100.0 - 1,200.0 = 45.2$. For year 4, apply $(1 + g)$ to both statements: $CI_4 = 145.2 \times 1.04 = 151.008$ and $BV_4 = 1,200.0 \times 1.04 = 1,248.0$, so $D_4 = 151.008 + 1,200.0 - 1,248.0 = 103.008$.

(b) Present value of D_1 to D_3 at 10%: $72.727 + 59.504 + 33.959 = 166.19$. Continuing value at $t = 3$: $103.008 / (0.10 - 0.04) = 1,716.80$, discounted three years = **1,289.86**. $V_0 = \text{€}1,456.05\text{m}$.

(c) Residual income: $RI_1 = 120.0 - (0.10 \times 1,000.0) = 20.0$; $RI_2 = 132.0 - 104.0 = 28.0$; $RI_3 = 145.2 - 110.0 = 35.2$; $RI_4 = 151.008 - (0.10 \times 1,200.0) = 31.008$. Present value of RI_1 to $RI_3 = 18.182 + 23.140$

+ 26.446 = **67.77**. Continuing value = $31.008/0.06 = 516.80$, discounted three years = **388.28**. $V_0 = 1,000.0 + 67.77 + 388.28 = \text{€}1,456.05\text{m}$ — identical to (b).

(d) The shortcut gives $D_4 = 45.2 \times 1.04 = 47.008$, a continuing value of $47.008/0.06 = 783.47$, discounted = 588.63, and $V_0 = 166.19 + 588.63 = \text{€}754.82\text{m}$ — an understatement of **48.2%**.

Economic meaning. The two models are the same model. The dividends model presents the value as a stream of distributions; the residual income model presents it as book value plus capitalised abnormal earnings — here, 1,000.0 of recorded capital plus 456.05 of expected wealth creation. Part (d) shows why the T+1 instruction matters: GAMMA's book equity grows at 9.1% in year 3 but at 4% thereafter, so the year 3 dividend is depressed by the year 3 retention and is not a valid base for growth. The correct terminal dividend is more than twice the shortcut figure.

Common mistakes. (i) Growing the terminal payoff instead of applying growth uniformly to the terminal-year statements. (ii) Charging required earnings on *closing* rather than opening book value — using $0.10 \times 1,040.0$ for RI_1 instead of $0.10 \times 1,000.0$. (iii) Forgetting to add back the opening book value in the residual income model. (iv) Concluding, when (b) and (c) agree, that the valuation is 'confirmed': the agreement confirms internal consistency, nothing more.

Exercise 17.4 — Terminal assumptions and the sensitivity of value

(a) Using $D_{(T+1)} = (1 + g)(CI_T - BV_T) + BV_T$ with $CI_T - BV_T = -11,029.8$: at $g = 2\%$, $D_{2026} = 1.02 \times (-11,029.8) + 107,296.4 = \textbf{96,046.0}$; at $g = 4\%$, $D_{2026} = 1.04 \times (-11,029.8) + 107,296.4 = \textbf{95,825.4}$. A *higher* growth rate produces a *lower* terminal dividend, because book equity exceeds comprehensive income and faster growth requires more retention. The effect on value is nonetheless strongly positive, because the denominator $(R_E - g)$ falls much faster than the numerator.

(b) Present value of the explicit dividends is 301,078.3 in both cases. At $g = 2\%$: continuing value = $96,046.0/0.04385 \times 0.733834 = 1,608,000$ approximately; total $\times 1.0319 / 17,352.1 = \113.49 (the whole chain is computed at the model's underlying rate of 6.385%). At $g = 3\%$: **\$141.59** (Table 17.17). A single percentage point of assumed perpetual growth is worth **\$28.10 per share, or 25%**.

(c) Terminal return on common equity of 92.4% exceeds the cost of equity of 6.39% by 86 percentage points, and the model capitalises that spread in perpetuity. This contradicts the terminal-period assumption unless one of its two escape clauses is invoked. The analyst must state which: either the spread is largely an **accounting distortion** — book equity depleted by cumulative repurchases and excluding internally generated intangibles — or it reflects a **durable departure from competitive equilibrium**. In the first case the economic spread is smaller than the measured one and the assumption is less aggressive than it appears; in the second, the durability of the competitive advantage is the central claim of the investment case and must be argued explicitly.

Economic meaning. Together, (b) and (c) locate the entire investment thesis in two numbers that no financial statement contains. This is not a defect of the method; it is the method being honest about where the judgement lies.

Common mistakes. (i) Assuming that a higher growth rate always raises the terminal payoff. (ii) Treating a high terminal return on equity as evidence of forecast quality rather than as a violation of the terminal-period principle requiring explanation. (iii) Reporting the base-case value without the grid.

Multiple-Choice Questions

MCQ 1. Correct answer: B. The PP&E roll-forward implies capital expenditure equal to depreciation, that is, a firm merely replacing its asset base while its volumes grow 12%. That is possible only if there is spare capacity, and the assumption must be documented. **A** is not an analytical objection but an assertion about a number whose plausibility depends on the firm and the industry. **C** is false: depreciation is a function of the asset base and its useful life, not of revenue. **D** overstates the rule — Table 17.4 specifies that PP&E tracks sales growth *except* growth arising from improved capacity utilisation or from price increases, which is precisely the exception at issue.

MCQ 2. Correct answer: C. This is the definition of the flexible financial account in Step 6, and the three responses listed are exactly those Table 17.5 lists for this direction of the imbalance. **A** is the most common misconception: the imbalance is the model's output, not a mistake. **B** confuses a balance sheet identity with an income statement outcome. **D** describes a response to the *opposite* imbalance — a surplus is what larger dividends absorb; reducing dividends would in fact be one way to address a requirement, but it is not among the adjustments listed and it does not follow from the sign as a matter of definition.

MCQ 3. Correct answer: B. The dividend is a difference between comprehensive income and the change in book value, and the two do not scale together unless book equity is already growing at exactly the long-run rate. Exercise 17.3(d) quantifies the error at 48%. **A** is true only in the special case where book equity grows at exactly g in year T . **C** is false: the explicit-horizon dividends are unaffected. **D** is false: the shortcut does not violate clean surplus, it simply fails to apply it to the terminal year.

MCQ 4. Correct answer: B. This is the structure of the two ladders. **A** is false: both are after-tax measures; what differs is the treatment of the *tax shield on interest*, which the entity measure removes by adding interest back on an after-tax basis. **C** is false: both deduct net capital expenditure. **D** reverses the matching rule — the entity flow is discounted at the WACC and the equity flow at the cost of equity.

MCQ 5. Correct answer: B. Every residual income term is zero, so $V_0 = BV_0$. **A** confuses residual income with value. **C** is a false contrast: the present value of the future dividends is *also* equal to book value in this case, since the three models are equivalent. **D** is wrong — a residual income of zero is a perfectly well-defined input, and it is precisely what the terminal-period assumption of competitive equilibrium implies.

MCQ 6. Correct answer: C. Multiples must be applied only after considering the firm's expected future in terms of profitability, risk and growth, and that fundamental characteristics be

considered before comparison with peers and industry averages. A firm whose book equity has been depleted by repurchases and whose principal assets are unrecognised has a book equity number that is not commensurable with that of the peers. **A** does not follow — a divergence between a shortcut and a full model is more naturally explained by the shortcut. **B** is unwarranted; the divergence is explicable without a market-efficiency claim. **D** overstates the case: multiples remain legitimate analytical and valuation tools when properly applied.

True or False

T/F 1. False. The first forecasting principle requires forecasts to be *unbiased and objective* — *neither conservative nor optimistic*. Risk is compensated through the discount rate; building a second, informal margin of safety into the numerator double-counts it and produces an expectation the analyst does not actually hold. The critical word is 'conservatively': prudence is a measurement convention, not a forecasting one.

T/F 2. False. The opposite is true: historical cash flows do *not* provide a good basis for projecting future cash flows, and the analyst should use the *implied* statement of cash flows computed from the projected income statements and balance sheets. The critical point is structural — the cash flow statement is a derivation from the other two, so forecasting it independently either duplicates or contradicts information already in the model.

T/F 3. False. Under clean surplus and an internally consistent forecast the three models are algebraically equivalent, so disagreement indicates one of three defects: incomplete or inconsistent forecasts, inconsistent WACC estimates, or incorrect continuing value computations. The critical word is 'range': the correct response is to locate the error, not to average it. Genuine valuation uncertainty is expressed through sensitivity analysis on the *inputs*, not through disagreement among models.

T/F 4. False. SG&A exhibits the sticky cost phenomenon: costs are relatively *less* responsive to reductions in sales growth than to increases. A symmetric rule therefore overstates the cost relief available in a downturn and understates the fall in operating income — an error that is invisible in the base case and decisive in the downside case that a credit analyst is employed to assess.

T/F 5. True only under conditions that are rarely met, and therefore false as stated. The market-to-book ratio reflects what market value *is*, not what it *should be*. Firms with off-balance-sheet assets and unrecognised shareholders' equity have structurally high MB ratios, and firms with higher MB ratios have been shown to earn higher subsequent returns on common equity. The overvaluation question can only be addressed by comparing MB with the **value-to-book** ratio derived from the analyst's own forecast of the spread between return on common equity and the cost of equity.

Exercise 17.5 — Reading a valuation you did not build (suggested solution)

(a) The agreement establishes that the pro forma statements articulate and that the continuing value has been computed consistently across the three specifications. It establishes nothing

whatever about whether the forecast is right. The three models are algebraically equivalent under clean surplus; applied to the same projection they *must* agree, and their agreement is a test the analyst has passed rather than evidence the committee should weigh.

(b) The discrepancy arises from the circularity in the WACC. The equity weight of 95.1% was computed from the market capitalisation of about \$2.11 trillion, while the model concludes that the equity is worth \$2.42 trillion. Using the model's own equity value would raise the equity weight, raise the WACC toward the cost of equity, and lower the entity value further. A second, smaller contributor is the deduction of debt at face rather than market value. The response is to iterate the weights or to use stated target weights — not to average \$141.59 and \$139.54.

(c) First, that the cost of equity is 6.39%; the sensitivity grid shows that at 8.00% the value falls to \$95.54 and the recommendation reverses. Second, that a terminal residual return on equity of 86 percentage points can be capitalised in perpetuity — an assumption that contradicts the terminal-period principle unless justified by a durable competitive advantage or identified as an accounting distortion.

(d) Three questions. First: at what cost of equity and at what growth rate does the model reproduce the market price of \$121.39, and is the difference from your estimates larger than the estimation error in beta and the market risk premium? Second: is the terminal return on equity an economic return or a measurement artefact produced by repurchase-depleted book equity and unrecognised intangibles — and if it is economic, what is the barrier to entry that sustains it? Third: the forecast assumes positive net debt issuance in every year to fund distributions exceeding comprehensive income; what happens to the value if that assumption is replaced by a self-funded distribution policy?

(e) The recommendation should be reformulated from a point estimate to a conditional statement: *the shares are attractive if the cost of equity is at or below approximately 6.9% at a 3% terminal growth rate, and if the firm's abnormal return on equity is durable rather than an artefact of a depleted book equity base; at a cost of equity of 8% the shares are unattractive on this forecast.* This is falsifiable, it identifies what would change the view, and it discloses to the committee the two judgements on which its capital is actually being risked.

Common mistakes. (i) Treating the agreement of three models as triangulation. (ii) Averaging the entity and equity results into a 'range'. (iii) Debating the near-term revenue forecast, which determines about 13% of the value, rather than the terminal assumptions, which determine the rest. (iv) Accepting a 17% upside as a margin of safety when the sensitivity grid shows the estimate moving by more than that in response to a 50-basis-point change in an unobservable input.

Chapter 18 — Monte Carlo Simulation for Discounted Cash Flow Valuation: From a Point Estimate of Value to a Distribution of Values

A. Numerical Exercises

Exercise 18.1 — A compact DCF and the leverage of the discount rate

Solution.

(a) Terminal value at the end of year 2: $TV_2 = FCFF_2 \times (1 + g) / (WACC - g) = 12.36 \times 1.02 / (0.09 - 0.02) = 12.607 / 0.07 = €180.1$ million. Enterprise value: $EV = 12.0/1.09 + (12.36 + 180.1)/1.09^2 = 11.01 + 192.46/1.1881 = 11.01 + 161.99 \approx €173.0$ million.

(b) The present value of the terminal value is $180.1/1.1881 = €151.6$ million, i.e. $151.6/173.0 \approx 88\%$ of enterprise value. With such a short explicit horizon, almost the whole valuation rests on the perpetuity — an extreme version of what Table 18.2 showed for Alpha Manufacturing (59%). This is why WACC and terminal growth rank among the dominant risk drivers in the tornado chart: they act on the largest component of value.

(c) With WACC = 10%: $TV_2 = 12.607/0.08 = €157.6$ m; $EV = 12.0/1.10 + (12.36 + 157.6)/1.21 = 10.91 + 140.5 \approx €151.4$ million (–12.5%). With WACC = 8%: $TV_2 = 12.607/0.06 = €210.1$ m; $EV = 12.0/1.08 + (12.36 + 210.1)/1.1664 = 11.11 + 190.7 \approx €201.9$ million (+16.7%). The same ± 1 percentage point moves value by –12.5% on one side and +16.7% on the other: the EV–WACC relationship is convex, because the discount rate enters the denominator of the perpetuity. Consequence for simulation: even if the WACC is drawn from a perfectly symmetric Normal distribution, the resulting distribution of enterprise values will be right-skewed — the mean will exceed the median, exactly the pattern visible in Table 18.8 (mean €106.3 m > median €105.8 m). Understanding this mechanical skew prevents the analyst from misreading it as “upside potential”.

Common mistakes: discounting the terminal value over one year instead of two (TV_2 is a value at the end of year 2); growing $FCFF_2$ by g twice; concluding from the convexity that a higher-volatility WACC “adds value” — the mean rises but so does downside dispersion, and decisions should weigh the whole distribution.

Exercise 18.2 — Probabilities from the simulated distribution

Solution.

(a) $z = (110 - 106.3)/9.4 = 0.39$. From the standard Normal distribution, $\Phi(0.39) \approx 0.652$, so $P(EV > 110) \approx 1 - 0.652 = 34.8\%$ ($\approx 35\%$) — roughly one simulated scenario in three. An acquirer paying €110 million is, in about two scenarios out of three, paying more than intrinsic value: the price can still be rational, but only if synergies or control benefits cover the gap.

(b) $z = (95 - 106.3)/9.4 = -1.20$; $\Phi(-1.20) \approx 0.115$, so $P(EV < 95) \approx 11.5\%$. If, for example, €95 million were the value threshold protecting the lenders in a leveraged transaction, the financing structure would be exposed in roughly one scenario out of nine — a number a credit committee can discuss, unlike a single point estimate.

(c) Normal 90% interval: $106.3 \pm 1.645 \times 9.4 = [\text{€}90.8 \text{ m}; \text{€}121.8 \text{ m}]$. The simulated interval is $[\text{€}91.7 \text{ m}; \text{€}121.8 \text{ m}]$. The upper bounds coincide, but the simulated lower percentile (91.7) sits above the Normal one (90.8): the empirical left tail is slightly thinner than Normal, and together with $\text{mean} > \text{median}$ this signals a mildly right-skewed distribution — consistent with the convexity mechanism of Exercise 18.1. The practical lesson: the Normal approximation is a useful cross-check, but decisions about tails (downside risk, VaR-type thresholds) should be read directly from the simulated percentiles, which is precisely what the simulation is for.

Common mistakes: using the median instead of the mean when standardizing; treating the Normal approximation as exact and quoting tail probabilities to a decimal place the model cannot support; comparing the asking price only with the mean and ignoring the probability statement.

Exercise 18.3 — Parameterizing a bounded variable: Beta-PERT versus Triangular

Solution.

(a) Beta-PERT: $\text{mean} = (15 + 4 \times 18 + 22)/6 = 109/6 = 18.17\%$; $\text{standard deviation} \approx (22 - 15)/6 = 1.17$ percentage points.

(b) Triangular: $\text{mean} = (15 + 18 + 22)/3 = 18.33\%$. $\text{Variance} = (15^2 + 18^2 + 22^2 - 15 \times 18 - 15 \times 22 - 18 \times 22)/18 = (225 + 324 + 484 - 270 - 330 - 396)/18 = 37/18 = 2.056$, hence $\text{standard deviation} = \sqrt{2.056} \approx 1.43$ percentage points.

(c) Both distributions are mildly right-skewed (the maximum is 4 points above the mode, the minimum only 3 below), so both means exceed the 18% mode. But the Triangular assigns a standard deviation about 23% larger than the PERT (1.43 vs. 1.17 points), because it spreads more probability toward the extremes, while the PERT concentrates mass around the most likely value. Using the exact PERT variance, $(\text{mean} - \text{min})(\text{max} - \text{mean})/7$, $\sigma = 1.32$ points, so the true gap is nearer 9%; the $(\text{max} - \text{min})/6$ rule overstates the concentration of the PERT. Fed into the simulation, the Triangular choice will mechanically widen the distribution of enterprise values and fatten both tails — with no new information about the business having been added. This is the concrete meaning of Section 18.3's warning: the parameterization of the inputs shapes the risk statistics of the output, so every distributional choice must be documented and defensible (Vose, 2008), not left as a software default.

Common mistakes: reading the mode (18%) as the expected value; using $(\text{max} - \text{min})/6$ as the standard deviation of the Triangular as well (it is a PERT approximation); forgetting that

widening an input distribution widens the output distribution even when the mean is unchanged.

B. Multiple Choice Questions

MCQ 1

Correct answer: (b).

(b) is correct: Sections 20.3.1 and 20.4 stress that the model itself is fixed; each iteration only re-samples the inputs and re-executes the same computation.

(a) is wrong — re-specifying the model at each iteration would make the results incomparable across scenarios.

(c) is wrong: the valuation perspective (FCFF discounted at WACC, throughout this chapter) is a design choice made once, before the simulation.

(d) is wrong: the horizon is part of the model structure, not a simulated variable.

MCQ 2

Correct answer: (c).

(c) is correct: the selection framework of Section 18.2.4 (Figure 18.4) simulates the upper-right quadrant — high uncertainty, high impact — as Table 18.3 does for Alpha Manufacturing.

(a) is wrong: data and software constrain implementation, but they do not define which variables matter economically.

(b) is wrong for the same reason — parsimony, not elegance, is the guiding principle.

(d) is wrong: WACC and terminal growth are simulated although they are not income statement items, while depreciation is not simulated although it is.

MCQ 3

Correct answer: (b).

(b) is correct: with positive correlation, good draws come together (high growth with high margins) and bad draws come together, stretching the tails; independence breaks this reinforcement, producing jointly inconsistent scenarios and an artificially narrow value distribution (Section 18.3.5; Hertz, 1964).

(a) is wrong: correlation affects the joint distribution, and hence the dispersion and tails of the output, even with unchanged means.

(c) is wrong: the first-order effect of omitting correlation is on dispersion, not on the expected value.

(d) is wrong: convergence concerns sampling stability (Section 18.5.5), not the correlation specification.

MCQ 4

Correct answer: (b).

(b) is correct, as stated in Section 18.4.3; modern software executes this range in seconds.

(a) is wrong: a hundred draws produce unstable statistics, especially in the tails.

(c) is wrong: beyond convergence, additional iterations refine extreme percentiles marginally but do not change the substance — precision improves only with the square root of the number of draws (Glasserman, 2004).

(d) is wrong: even perfectly specified distributions are sampled with error; too few draws leave that sampling error material.

MCQ 5

Correct answer: (c).

(c) is correct: the CDF maps any threshold into a probability directly (Section 18.5.2, Figure 18.10; Table 18.9).

(a) is wrong: the tornado chart ranks the drivers of valuation risk; it says nothing about the probability of a specific outcome.

(b) is wrong: convergence analysis checks the statistical stability of the results.

(d) is wrong: the correlation matrix is an input of the model, not an output describing enterprise value.

MCQ 6

Correct answer: (b).

(b) is correct: Section 18.3.3 motivates the Beta exactly on boundedness and asymmetry, with a smoother density than the Triangular's artificial peak.

(a) is wrong: all the distributions of Table 18.4 are routinely available in simulation software.

(c) is wrong: flexibility — symmetric, right- or left-skewed, nearly uniform — is the Beta's defining feature, not symmetry.

(d) is wrong: the Beta-PERT still requires three parameters (minimum, most likely, maximum), and Exercise 18.3 shows how consequential they are.

C. True/False Questions

True/False 1

FALSE. The critical word is “replaces”: the simulation is an extension of the DCF framework — the valuation model is unchanged and is simply re-executed under alternative inputs (Sections 20.4 and 20.4.2). It enriches the point estimate with a distribution; it does not produce value by a different logic.

True/False 2

TRUE. From Table 18.2: €62.2 million of €105.0 million, about 59%. This is why WACC and terminal growth — the two variables governing the perpetuity — emerge among the dominant risk drivers in the tornado chart (Section 18.5.4).

True/False 3

FALSE. The critical part is “will typically change”: convergence means precisely that summary statistics have stabilized, so further iterations refine extreme percentiles at most marginally (Sections 20.4.3 and 20.5.5).

True/False 4

FALSE. The critical claim is “compensate”: simulation magnifies both the strengths and the weaknesses of the underlying model (Practical Insight, Section 18.4). Guideline 1 of Section 18.6 therefore requires validating the deterministic model before introducing uncertainty.

True/False 5

TRUE. Correlations preserve economically meaningful relationships (strong demand with better margins; weakness hitting sales and profitability together), preventing jointly inconsistent draws and correcting the artificially narrow tails of independence-based models (Section 18.3.5).

True/False 6

FALSE. The critical part is “with the mean only”: the point of the simulation is to move from a comparison of two numbers to a probability statement — the probability that intrinsic value exceeds the price, lower-tail percentiles, VaR-type thresholds (Section 18.5.3 and Exercise 18.2).

D. Real-World Interpretation Exercise

Exercise 18.4 — An acquisition memo for Alpha Manufacturing

Solution (guided).

(a) Valuation judgment. The asking price of €110 million lies above both the mean (€106.3 m) and the median (€105.8 m) of the simulated distribution. From Exercise 18.2, the probability that intrinsic value exceeds the price is only about 35%: in roughly two scenarios out of three the fund would be paying more than stand-alone value. The memo should therefore state that the price is not justified on a stand-alone basis and quantify the value creation (synergies, operating improvements, multiple expansion) needed to close the gap — about €4 million, the distance between the price and the central estimates (€3.7m vs. the mean, €4.2m vs. the median) — before any margin of safety.

(b) Financing risk. The covenant threshold of €95 million is breached in about 11–12% of scenarios (Exercise 18.2(b); the simulated 5th percentile is €91.7 m, so the breach zone extends well into the observed range). One scenario in nine is not a remote tail: the memo should either recommend a lower debt quantum, covenant headroom re-negotiation, or explicit contingency planning for the breach scenarios — this is downside-risk language a credit committee can act on, unavailable from the deterministic €105 million alone.

(c) Due diligence priorities. The tornado chart ranks revenue growth and WACC as the dominant sources of valuation risk, followed by EBITDA margin and terminal growth. Due diligence effort should be allocated accordingly: order backlog, customer concentration and the demand outlook of the automotive and machinery end-markets (revenue growth); financing structure, rate hedging and the cost of the acquisition debt (WACC); the credibility of the 15–22% margin bounds elicited from management (Exercise 18.3). Working capital and capex, lower in the ranking, deserve verification but not the committee's scarce attention.

(d) Caveats. The memo must disclose that the distribution reflects the modeling choices documented in Tables 20.4–20.6 — distribution families, parameters, and the correlation matrix — and that these were selected by judgment supported by evidence, not observed; that the mild right skew of the output is partly mechanical (the convexity of value in the discount rate, Exercise 18.1); and that convergence was verified before quoting the statistics (Figure 18.13). A simulated valuation is an argument, not an oracle: its authority comes entirely from the transparency of its assumptions.

Common mistakes: quoting the probability of overpaying without stating the stand-alone perimeter (synergies change the distribution, not just the price test); reading the 5th percentile as a “worst case” (5% of scenarios lie below it); allocating due diligence uniformly across all drivers despite the tornado ranking.